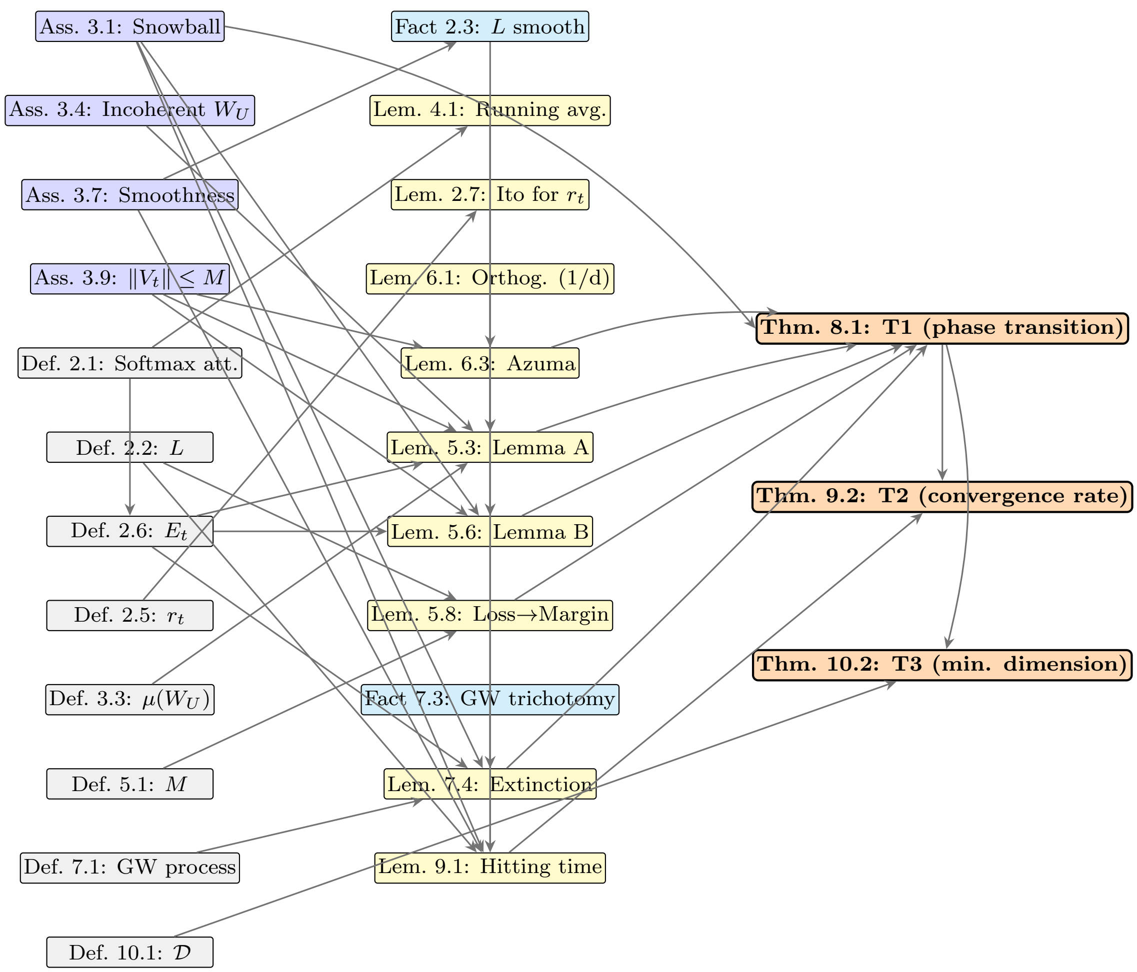
High-dimensional random walks plus rare effective tokens: a phase-transition theory of reasoning convergence

1 Roadmap: dependency graph of the formal results

The figure below visualises the logical dependencies among the assumptions, definitions, facts, lemmas, and theorems of this paper. An arrow $A \to B$ indicates that A is used in the statement or proof of B.



How to read the graph. The left column collects all *assumptions* (blue) and *definitions* (gray) on which the analysis depends. The middle column contains the *facts* (cyan) and *lemmas* (yellow), arranged in approximately top-to-bottom topological order. The right column contains the three *theorems* (orange). The proof of each downstream object requires every upstream object reachable via the directed arrows; the graph therefore doubles as a checklist for proof completeness and as a guide to the order of reading.

The two "hub" lemmas in the middle column are Lemma 5.3 (Lemma A, the incorrect-logit max bound) and Lemma 5.6 (Lemma B, the correct-logit signal lower bound); together with the loss-to-margin bridge Lemma 5.8 they support Theorem 8.1 (the headline phase-transition theorem). The remaining dependencies are auxiliary: Lemma 7.4 discharges the extinction branch of T1 via Galton-Watson coupling, Lemma 9.1 converts T1's success into the T2 convergence rate, and the definition of $\mathcal{D}$ inverts the T1 threshold to give T3.

2 Preliminaries

2.1 Notation summary

For ease of cross-reference, the table below lists every mathematical symbol used in the paper, with a one-line description and a pointer to its first formal introduction.

Symbol	Description	First in
$\mathcal{V}$	Vocabulary (token alphabet)	Section 2.3
$\lvert\mathcal{V}\rvert$	Vocabulary size	Section 2.3
n	Answer length (sentences treated atomically in $\mathcal{V}^n$)	Section 2.4
$\lvert\mathcal{V}\rvert^n$	Effective alphabet size at the answer-event level	Section 2.4
$\mathcal{A}(Q) \subseteq \mathcal{V}^n$	Verifier-accepted answer set (sequences)	Section 2.4
a, a'	Answer (sequence) iterator, $a, a' \in \mathcal{V}^n$	Section 2.3
$a^\star$	Correct answer of max sequence-embedding norm	Lemma 5.6
d	Residual-stream (hidden) dimension	Section 2.2
Q	Question (held fixed throughout)	Section 2.2
$x_t \in \mathbb{R}^d$	Attention output (residual state) at step t	Definition 2.1
$V_t \in \mathbb{R}^d$	Attention value vector at step t	Definition 2.1
$g_t = x_t - x_{t-1}$	Per-step increment	Section 2.5
s_j	Cumulative softmax denominator	Definition 2.1
$w_{T,t} \in (0, 1]$	Softmax weight ($\sum_t w_{T,t} = 1$)	Definition 2.1
$\tau = 1/T$	Average per-step softmax weight	Eq. (18)
$W_U \in \mathbb{R}^{\lvert\mathcal{V}\rvert^n \times d}$	Sequence-unembedding matrix (abstract, $\lvert\mathcal{V}\rvert^n$ rows)	Section 2.3
W_U^a	Sequence-embedding row of $a \in \mathcal{V}^n$	Section 2.3
$p = \mathrm{softmax}(W_U x)$	Softmax probability vector over $\mathcal{V}^n$	Remark 5.2
q_C	$\mathcal{A}$-conditional posterior	Remark 5.2
$\pi(x; Q)$	Correct-answer mass	Eq. (2)
$L(x; Q)$	Constrained-softmax loss $-\log \pi(x; Q)$	Definition 2.2
$L_0 = L(x_0; Q)$	Initial loss (conditioning value)	Lemma 9.1
L_{sm}	Smoothness constant of L	Fact 2.3
$r_t = \lVert x_t \rVert_2$	Radial coordinate	Definition 2.5
$\mathcal{F}_t$	Filtration $\sigma(x_0, \ldots, x_t, E_1, \ldots, E_t)$	Section 2.6
$E_t \in \{0, 1\}$	Effective-token indicator	Definition 2.6
$\lambda(L)$	State-dependent effective-token rate	Assumption 3.1
$\lambda_0(Q)$	Initial / snowball-region effective rate	Assumption 3.1
$L^\star(Q)$	Snowball-region cutoff loss	Assumption 3.1
$\mu(W_U)$	Row-incoherence of W_U	Definition 3.3
μ_0, R_U	Incoherence and row-norm bounds	Assumption 3.4
M	Value-norm bound $\lVert V_t \rVert_2 \le M$	Assumption 3.9
σ	Per-step noise scale (radial-Ito)	Lemma 2.7
$M(x; Q)$	Logit margin (verifier-success quantity)	Definition 5.1
$L_{\mathrm{dec}}(\lvert\mathcal{V}\rvert^n, \lvert\mathcal{A}\rvert)$	Decode-threshold loss	Lemma 5.8
$\kappa(\mu_0) = 1 + \mu_0$	Lemma A drift-absorption constant	Theorem 8.1
$\cos\theta_0 \in (0, 1]$	Effective-step alignment cosine	Eq. (17)
c_1, c_2	T1 / T2 prefactors	Theorem 8.1
λ_{c}	Critical effective-token rate	Theorem 8.1
$\mathcal{S}$	Snowball event (verifier success in window)	Section 7
Z_n, m, Z	GW generation size, offspring mean, total progeny	Definition 7.1
T_{dec}, T_{Ldec}	Margin / loss-target hitting times	Lemma 9.1
$T_{\max}$, $T_\star$	Trajectory horizon, T2 high-prob horizon	Theorem 8.1
$\mathcal{D}(Q)$	Minimum reasoning dimension	Definition 10.1
$c, C, C_1, C_2, \ldots$	Universal positive constants	Section 2
$\delta, \delta_+, \delta_-$	Failure probabilities in $1 - \delta$ statements	Theorem 8.1

We introduce the architectural recurrence, the loss objective, the radial summary statistic that drives the analysis, and the Ito-style expansion that connects radial increments to per-step noise variance. Throughout, c, C, C_1, $C_2, \ldots$ denote universal positive constants whose values may change from line to line and depend only on the quantities specified in each lemma's hypothesis.

2.2 Single-head softmax attention recurrence

Fix a question Q and a residual-stream dimension $d \in \mathbb{N}$. Throughout, $\langle q, k_i \rangle$ abbreviates the scaled inner product $q^\top k_i/\sqrt{d_k}$ of [11].

Definition 2.1 (Softmax attention with cumulative running sum)**.** For a query $q \in \mathbb{R}^d$ and key-value pairs $\{(k_i, V_i)\}_{i=1}^{j} \subset \mathbb{R}^d \times \mathbb{R}^d$, the cumulative softmax output at step $j \geq 1$ is

$$x_j \;=\; \frac{1}{s_j} \sum_{k=1}^{j} e^{\langle q, k_k \rangle}\, V_k, \qquad s_j \;\coloneqq\; \sum_{i=1}^{j} e^{\langle q, k_i \rangle}. \tag{1}$$

We adopt the conventions $x_0 \coloneqq 0$ and $s_0 \coloneqq 0$. The per-step convex weights are $w_{j,k} \coloneqq e^{\langle q, k_k \rangle}/s_j \in (0, 1]$ with $\sum_{k=1}^{j} w_{j,k} = 1$.

Eq. (1) is the standard scaled dot-product attention of [11] written as a prefix-cumulative sum; this presentation makes the running-average recurrence (Lemma 4.1) immediate. The single-head simplification suffices for the analysis below: multi-head extensions add only direct-sum bookkeeping and do not alter the high-dimensional concentration arguments.

2.3 Constrained-softmax loss

Let $W_U \in \mathbb{R}^{|\mathcal{V}|^n \times d}$ be the (abstract) *sequence-unembedding* matrix, whose rows W_U^a are indexed by answer sequences $a \in \mathcal{V}^n$ (cf. Section 2.4 below for the answer set $\mathcal{A}(Q) \subseteq \mathcal{V}^n$ and Remark 2.4 for the linearisation interpretation of the abstract W_U). The correct-mass functional is

$$\pi(x; Q) \;\coloneqq\; \sum_{a \in \mathcal{A}(Q)} \operatorname{softmax}(W_U x)_a \;\in\; (0, 1], \tag{2}$$

where the softmax is taken over the effective alphabet $\mathcal{V}^n$ of size $|\mathcal{V}|^n$.

Definition 2.2 (Constrained-softmax loss)**.** The constrained-softmax loss along the trajectory at residual state $x \in \mathbb{R}^d$ is

$$L(x; Q) \;\coloneqq\; -\log \pi(x; Q) \;\in\; [0, \infty). \tag{3}$$

Fact 2.3 (Smoothness of L)**.** The loss $L(\cdot; Q)$ is twice continuously differentiable on $\mathbb{R}^d$ and its Hessian satisfies $\left\|\nabla^2 L(x; Q)\right\|_{\mathrm{op}} \leq \|W_U\|_{\mathrm{op}}^2/2$ uniformly in x; in particular $L(\cdot; Q)$ is L_{sm}-smooth with $L_{\mathrm{sm}} \leq \|W_U\|_{\mathrm{op}}^2/2$.

Fact 2.3 is the standard log-sum-exp Hessian bound $\left\|\nabla^2(-\log \sum e^{a(x)})\right\|_{\mathrm{op}} \leq \frac{1}{4} \sup_a \|\nabla a\|_2^2 \cdot 2 = \|W_U\|_{\mathrm{op}}^2/2$ specialised to a linear $a(x) = (W_U x)_a$ over $a \in \mathcal{V}^n$; no separate proof is needed.

2.4 Answer set and decoding

Fix a question Q. Let $n \in \mathbb{N}$ denote the answer length (a hyperparameter of the benchmark; e.g., $n = 1$ for single-token answers, $n \approx 10$ for short-form math). The *answer set* is

$$\mathcal{A}(Q) \;\subseteq\; \mathcal{V}^n, \tag{4}$$

the set of token sequences $(a_1, \ldots, a_n) \in \mathcal{V}^n$ that the verifier accepts as correct.

Sequence-level decoding event. The framework analyses the model's decoding decision at the position immediately following the `</think>` token as a *single atomic answer event*: the entire sequence $a = (a_1, \ldots, a_n) \in \mathcal{V}^n$ produced by the model is the unit of analysis, not the per-token argmax. We abstract this commitment as an argmax over the effective alphabet $\mathcal{V}^n$ of size $|\mathcal{V}|^n$: given the residual state x_T at `</think>`+1, the decoded answer is

$$a^*(x_T) \;\coloneqq\; \arg\max_{a \in \mathcal{V}^n} (W_U\, x_T)_a, \tag{5}$$

where $W_U \in \mathbb{R}^{|\mathcal{V}|^n \times d}$ is the abstract sequence-unembedding matrix of Section 2.3. The verifier accepts iff

$$a^*(x_T) \;\in\; \mathcal{A}(Q). \tag{6}$$

Verifier success at horizon T is the event $\{a^*(x_T) \in \mathcal{A}(Q)\}$, and the total number of *incorrect* answer sequences against which the correct ones must compete is $|\mathcal{V}|^n - |\mathcal{A}(Q)|$.

Remark 2.4 (Sequence-level linearisation of the autoregressive decoder). In a real autoregressive LLM, decoding at `</think>`+1 is per-token, not per-sequence: the model emits a_1 by token-level argmax on the next-token logits, then a_2 conditional on (x_T, a_1), and so on. The abstract matrix $W_U \in \mathbb{R}^{|\mathcal{V}|^n \times d}$ with one row per answer sequence $a \in \mathcal{V}^n$ is a *linearisation*: we treat the joint sequence log-probability $\log P(a \mid x_T)$ as approximately linear in x_T with effective sequence-embedding $W_U^a \in \mathbb{R}^d$, and absorb the autoregressive multi-step composition into the abstract row W_U^a. The bounded-incoherence assumption (Assumption 3.4 below) is therefore applied to the sequence-embedding rows $\{W_U^a\}_{a \in \mathcal{V}^n}$ of this abstract W_U, not to the per-token unembedding rows of the real LLM. This linearisation is the working stylisation of the framework: it treats the entire `</think>`+1 answer event as a single decoding event against an effective alphabet of size $|\mathcal{V}|^n$. The cost in difficulty is captured by the $\log(|\mathcal{V}|^n / |\mathcal{A}|) = n \log |\mathcal{V}| - \log |\mathcal{A}|$ factor that appears throughout the analysis, showing that difficulty grows linearly with answer length n.

In this notation, the correct-mass functional Eq. (2) sums the softmax probabilities of the answer set $\mathcal{A}(Q)$, and the constrained-softmax loss Eq. (3) measures $L(x; Q) = -\log \mathbb{P}[\, \arg\max_{a \in \mathcal{V}^n} (W_U x)_a \in \mathcal{A}(Q)\,]$ in the limit of soft argmax decoding. By Eqs. (5) and (6), the entire phase-transition analysis of Section 8 bounds the verifier-success probability *exactly* (no separate per-token-to-sequence bridge is required, because the sequence-level argmax is the verifier's operational quantity by construction).

2.5 Radial coordinate and trajectory

The Hidden-Markov two-mode decomposition formalises the per-step increment $g_t \coloneqq x_t - x_{t-1}$ in terms of an effective component and a noise component. The per-step increment satisfies $g_t = \tau \cdot V_t$ with τ the softmax weight of the most recent value vector; on effective steps ($E_t = 1$) V_t has a deterministic component aligned with the negative loss gradient, while on noise steps ($E_t = 0$) V_t is mean-zero conditional on $\mathcal{F}_{t-1}$. The per-step formalism is invoked in Section 5 (signal accumulation) and Section 6 (cumulative-noise concentration); the radial coordinate below is a one-dimensional summary statistic retained from the v1 framework for the regularity calculations of Lemma 2.7.

Definition 2.5 (Radial coordinate)**.** For the trajectory $(x_t)_{t \geq 0}$ of Definition 2.1, the *radial coordinate* is

$$r_t \;\coloneqq\; \|x_t\|, \qquad t \geq 0.$$

The choice $r_t = \|x_t\|$ is the simplest summary statistic that co-varies monotonically with the loss $L(x_t;Q)$ in the near-target regime (where π is dominated by the inner product $\langle x, W_U^\top \mathbf{1}_{\mathcal{A}}\rangle$ and so by $\|x_t\|$ up to direction). The reduction to a scalar process is the working stylisation of this paper; the full vector-valued analysis recovers the same scaling on r_t alone (cf. the order-parameter framework of [9], where high-dimensional SGD dynamics close onto low-dimensional summary statistics under the critical step-size scaling that we adopt in Section 3).

2.6 Filtration and effective indicator

Let $(\mathcal{F}_t)_{t\geq 0}$ be the filtration generated by $(x_0, x_1, \ldots, x_t)$ together with the indicator history introduced below.

Definition 2.6 (Effective-token indicator)**.** The *effective-token indicator* is a sequence $(E_t)_{t\geq 1}$ of $\{0,1\}$-valued random variables, each $\mathcal{F}_t$-measurable, with conditional distribution

$$\mathbb{P}\big[\, E_t = 1 \;\big|\; \mathcal{F}_{t-1} \,\big] \;=\; \lambda\big(L(x_{t-1};Q)\big), \tag{7}$$

where $\lambda : \mathbb{R}_{\geq 0} \to [0,1]$ is a measurable function fixed by the model-and-question pair (model, Q); the structural constraints on λ are stated as Assumption 3.1 in Section 3.

The effective indicator partitions per-step increments into two populations: when $E_t = 1$ the increment contains a population-level drift component aligned with $-\nabla L(x_{t-1};Q)$, and when $E_t = 0$ the increment is pure spherical noise. The two-mode decomposition is invoked in Lemma 5.6 (signal contribution under the snowball assumption) and Lemma 6.3 (noise concentration).

2.7 An Ito-style expansion for the radial coordinate

The map $x \mapsto \|x\|$ is smooth away from the origin, with gradient $\nabla \|x\| = x/\|x\|$ and Hessian $\nabla^2 \|x\| = (I - xx^\top/\|x\|^2)/\|x\|$. A second-order Taylor expansion of $\|x+g\|$ in the increment g gives the centrifugal correction term that drives the random-walk phase of the dynamics.

Lemma 2.7 (Discrete Ito expansion for the radial coordinate)**.** *Suppose $x \in \mathbb{R}^d$ with $\|x\| = r > 0$ and let $g \in \mathbb{R}^d$ be a mean-zero random vector with covariance $\mathbb{E}[gg^\top] = (\sigma^2/d)\cdot I$ and $\mathbb{E}\|g\|^3 \leq \gamma_3\sigma^3$ for some finite γ_3. Then*

$$\mathbb{E}\big[\, \|x+g\| - r \,\big] \;=\; \frac{(d-1)\,\sigma^2}{2\,r\,d} \;+\; R(r,\sigma,d), \qquad |R(r,\sigma,d)| \;\leq\; \frac{C\gamma_3\sigma^3}{r^2}, \tag{8}$$

for an absolute constant $C > 0$, provided $\sigma \leq c\,r\sqrt{d}$ for an absolute constant $c \in (0,1)$.

Proof. The proof proceeds in two stages: a second-order Taylor expansion of $\|\cdot\|$ around x, then an explicit computation of the covariance trace on the constant-coefficient quadratic term.

Step 1: Taylor expansion. Write $f(y) \coloneqq \|y\|$ and expand $f(x+g) - f(x)$ as a second-order Taylor series with integral remainder:

$$f(x+g) - f(x) \;=\; \langle \nabla f(x), g\rangle \;+\; \frac{1}{2}\, g^\top \nabla^2 f(x)\, g \;+\; R_3(x,g),$$

where the remainder satisfies the third-order bound $|R_3(x,g)| \leq \frac{1}{6}\sup_{\xi\in[x,x+g]} \big\|\nabla^3 f(\xi)\big\|_{\mathrm{op}} \cdot \|g\|^3$. On the ball $\|y - x\| \leq r/2$ the operator norm of the third derivative of $\|\cdot\|$ is bounded by C/r^2 for an absolute C (by direct computation from the explicit form of $\nabla^k \|y\|$, $k \leq 3$), so $|R_3(x,g)| \leq C\|g\|^3/r^2$ when $\|g\| \leq r/2$. The hypothesis $\sigma \leq cr\sqrt{d}$ with c small enough makes the event $\|g\| > r/2$ negligible

after truncation, with truncation error absorbed into the R-term of Eq. (8) (the truncated and original expectations differ by at most $\mathbb{E}[\|g\| \cdot \mathbf{1}\{\|g\| > r/2\}] \le \mathbb{E}\,\|g\|^3/(r/2)^2 \le 4\gamma_3\sigma^3/r^2$).
Step 2: expectation of the Taylor terms. Taking expectations and using $\mathbb{E}g = 0$:

$$\begin{aligned}
\mathbb{E}\,\langle \nabla f(x), g\rangle \;&=\; \langle \nabla f(x), \mathbb{E}g\rangle \;=\; 0,\\
\mathbb{E}\big[\, g^\top \nabla^2 f(x)\, g \,\big] \;&=\; \operatorname{tr}\big(\nabla^2 f(x) \cdot \mathbb{E}[gg^\top]\big) \;=\; \frac{\sigma^2}{d}\operatorname{tr}\big(\nabla^2 f(x)\big).
\end{aligned}$$

The Hessian of $\|\cdot\|$ at x with $\|x\| = r > 0$ is $\nabla^2\|x\| = \frac{1}{r}\big(I - \frac{xx^\top}{r^2}\big)$, whose trace is $\frac{1}{r}(d-1)$. Combining,

$$\frac{1}{2}\,\mathbb{E}\big[\, g^\top \nabla^2 f(x)\, g \,\big] \;=\; \frac{\sigma^2}{2d} \cdot \frac{d-1}{r} \;=\; \frac{(d-1)\sigma^2}{2rd}.$$

Substituting Steps 1 and 2 yields Eq. (8), with $R(r, \sigma, d) \coloneqq \mathbb{E}[R_3(x, g)] +$ truncation error and the stated $C\gamma_3\sigma^3/r^2$ bound. This completes the proof. □

Lemma 2.7 is the discrete analogue of the Ito expansion of $d\,\|X_t\|$ for an isotropic diffusion: the leading $\sigma^2/(2r)$ term is the *centrifugal* drift that pushes the radial coordinate outward in the absence of a counter-acting effective drift, and the $(1 - 1/d)$ factor is the finite-d correction. In the v3 framework of this paper, the $\Theta(1/\sqrt{d})$ scaling of the critical effective-token rate does *not* arise from this radial calculation; instead it is derived from the row-incoherence of the unembedding matrix W_U via Lemma 5.3 in Section 5. Lemma 2.7 is retained for the regularity calculations in the extinction branch of Theorem 8.1 where the trajectory escapes the snowball region and the radial coordinate may drift outward.

3 Assumptions

The framework rests on a single load-bearing behavioural assumption (the *snowball assumption*, Assumption 3.1) and a single geometric assumption on the unembedding matrix (*bounded incoherence*, Assumption 3.4). Two auxiliary regularity conditions on the loss and the value vectors are stated for completeness.

3.1 The snowball assumption

Assumption 3.1 (Snowball / state-dependent effective rate)**.** There exist a measurable function $\lambda : \mathbb{R}_{\ge 0} \to [0, 1]$, a question-dependent threshold $L^\star(Q) > 0$, and a snowball-rate constant $\lambda_0(Q) > 0$ such that the effective-token indicator of Definition 2.6 satisfies

$$\mathbb{P}\big[\, E_t = 1 \;\big|\; \mathcal{F}_{t-1} \,\big] \;=\; \lambda\big(L(x_{t-1}; Q)\big),$$

with the cutoff structure

$$\lambda(L) \;\ge\; \lambda_0(Q) \cdot \mathbf{1}\{L < L^\star(Q)\}, \qquad \lambda(L) \;\le\; \overline{\lambda} \cdot \mathbf{1}\{L < L^\star(Q)\},$$

for some $\overline{\lambda} \in [\lambda_0(Q), 1]$.

Remark 3.2 (Mildness of Assumption 3.1). Assumption 3.1 is the only behavioural assumption on the reasoning trajectory. It is strictly milder than a per-token gradient-correctness condition in three concrete senses. First, no uniform lower bound on $\lambda(L)$ across all L is required: the rate is permitted to vanish in bad regions $L > L^\star(Q)$, which allows the analysis to distinguish trajectories that fall outside the basin from those that begin inside it. Second, no per-token alignment condition is imposed; λ is a *rate*, not a per-step gradient-direction requirement. Third, no anchor set or sub-token-vocabulary structure is required; the effective indicator is intrinsic to the loss process. The contemporary references [7, 4, 8, 13] document that modern reasoning-tuned LLMs emit explicit intermediate tokens during inference; Assumption 3.1 models the proportion of such tokens that materially contribute to loss decrease via the state-dependent rate λ.

3.2 Bounded-incoherence sequence-unembedding

The phase-transition scaling in d emerges from a geometric property of the (abstract) sequence-unembedding matrix $W_U \in \mathbb{R}^{|\mathcal{V}|^n \times d}$: rows representing distinct answer sequences are nearly orthogonal. This is captured by a single deterministic parameter, the incoherence $\mu(W_U)$. Per Remark 2.4, W_U is the abstract linearisation of the autoregressive joint sequence log-probability; the incoherence below is applied to its sequence-embedding rows $\{W_U^a\}_{a\in\mathcal{V}^n}$, not to the per-token unembedding rows of the real LLM.

Definition 3.3 (Row-incoherence of a sequence-unembedding matrix)**.** For a sequence-unembedding matrix $W_U \in \mathbb{R}^{|\mathcal{V}|^n \times d}$ with rows $W_U^a \in \mathbb{R}^d$ indexed by answer sequences $a \in \mathcal{V}^n$, the *row-incoherence* is

$$\mu(W_U) \;:=\; \max_{a\neq a'\in\mathcal{V}^n} \frac{|\left\langle W_U^a, W_U^{a'}\right\rangle|}{\|W_U^a\|_2 \cdot \|W_U^{a'}\|_2}. \tag{9}$$

Assumption 3.4 (Bounded incoherence and row norms)**.** There exist constants $\mu_0 \in [0, 1/2]$ and $R_U \in [1, \infty)$ such that the sequence-unembedding matrix satisfies

$$\mu(W_U) \;\le\; \mu_0, \qquad \max_{a\in\mathcal{V}^n} \|W_U^a\|_2 \;\le\; R_U. \tag{10}$$

Remark 3.5 (Interpretation and realism of Assumption 3.4)*.* The bound $\mu(W_U) \le \mu_0$ formalises "answer-sequence embeddings are not too colinear" in the abstract sequence-unembedding of Remark 2.4. For an idealised W_U with i.i.d. rows drawn uniformly from S^{d-1} (one per answer sequence), the typical incoherence is $\mu(W_U) = O\big(\sqrt{n\log|\mathcal{V}|/d}\big) = O\big(\sqrt{\log(|\mathcal{V}|^n)/d}\big)$ by standard concentration ([12], Chapter 3); the bound also holds deterministically for sensibly normalised linearisations whose sequence-embedding similarity is bounded away from ± 1. The row-norm bound $R_U \ge \max_a \|W_U^a\|$ controls the scale of sequence-logits and is trivially $R_U = 1$ when rows are unit-normalised. *Note*: we do *not* assume any random-matrix model for W_U. The assumption is deterministic, and the linearisation of Remark 2.4 fixes the abstract W_U once the policy is fixed.

Remark 3.6 (Geometric vs. dynamical formulation of the d-scaling)*.* Assumption 3.4 is a geometric property of W_U, not a dynamical postulate on the trajectory. The $\Theta(1/\sqrt{d})$ scaling of the critical effective-token rate emerges as a *derived consequence* via Lemma 5.3 (in Section 5), applied to the competition between $|\mathcal{V}|^n - |\mathcal{A}(Q)|$ incorrect-answer-sequence projections. The dimensional dependence enters through the sub-Gaussian variance proxy $1/d$ of the projection $\left\langle W_U^{a'}, x_t\right\rangle$ (Lemma A), not through a postulated step-size rule on $\|\nabla L\|$.

3.3 Auxiliary regularity assumptions

The two assumptions in this subsection are standard regularity conditions used to keep the per-step expansions well-defined. Neither is load-bearing for the phase-transition statement.

Assumption 3.7 (Bounded smoothness)**.** The constrained-softmax loss $L(\cdot; Q) : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ of Definition 2.2 is L_{sm}-smooth, with $L_{\text{sm}} \leq \|W_U\|_{\text{op}}^2 /2$, where $\|W_U\|_{\text{op}}$ denotes the operator (spectral) norm.

Remark 3.8 (Smoothness is automatic from the architecture)*.* Assumption 3.7 is not an additional postulate: it restates the standard Hessian bound for the log-sum-exp loss (Fact 2.3). It is named as an assumption only to make the dependency tracking explicit in subsequent statements.

Assumption 3.9 (Bounded value norms)**.** There exists $M > 0$ such that the per-step value vectors of Definition 2.1 satisfy $\|V_j\|_2 \leq M$ almost surely for every $j \geq 1$.

Remark 3.10 (Realism of the value-norm bound)*.* Assumption 3.9 is standard for analyses of attention trajectories and is satisfied by any pretrained transformer with layer-norm-bounded activations; see [11] for the architectural specification and [12] (Chapter 3) for the high-dimensional concentration arguments that make M depend only weakly on d in practice.

4 The softmax recurrence as a running weighted average

Lemma 4.1 (Softmax recurrence as running average)**.** *Let x_j, $w_{j,k}$, and s_j be defined as in Definition 2.1. With the convention $x_0 := 0$ and $s_0 := 0$, the sequence $(x_j)_{j\geq 0}$ satisfies, for every $j \geq 1$, both the recurrence*

$$x_j \;=\; \frac{s_{j-1}}{s_j}\, x_{j-1} \;+\; \frac{e^{\langle q,k_j\rangle}}{s_j}\, V_j, \tag{11}$$

and the convex-combination identity

$$x_j \;=\; \sum_{k=1}^{j} w_{j,k}\, V_k, \qquad w_{j,k} \;\geq\; 0, \qquad \sum_{k=1}^{j} w_{j,k} \;=\; 1. \tag{12}$$

Proof. The proof is a one-line algebraic identity followed by induction on j.
Step 1: recurrence. By the definition $s_j = \sum_{i=1}^{j} e^{\langle q,k_i\rangle}$, we have $s_j = s_{j-1} + e^{\langle q,k_j\rangle}$ for every $j \geq 1$. Multiplying Eq. (1) by s_j gives

$$s_j\, x_j \;=\; \sum_{k=1}^{j} e^{\langle q,k_k\rangle}\, V_k \;=\; \sum_{k=1}^{j-1} e^{\langle q,k_k\rangle}\, V_k \;+\; e^{\langle q,k_j\rangle}\, V_j \;=\; s_{j-1}\, x_{j-1} \;+\; e^{\langle q,k_j\rangle}\, V_j,$$

where the last equality used $s_{j-1}x_{j-1} = \sum_{k=1}^{j-1} e^{\langle q,k_k\rangle} V_k$ from Eq. (1) at index $j-1$ (and the conventions $x_0 = 0$, $s_0 = 0$ from the lemma statement). Dividing both sides by $s_j > 0$ yields Eq. (11).
Step 2: convex weights. The weights $w_{j,k} = e^{\langle q,k_k\rangle}/s_j$ are non-negative since exponentials are positive and $s_j > 0$. Their row sum is

$$\sum_{k=1}^{j} w_{j,k} \;=\; \frac{1}{s_j} \sum_{k=1}^{j} e^{\langle q,k_k\rangle} \;=\; \frac{s_j}{s_j} \;=\; 1,$$

which establishes Eq. (12) directly from Eq. (1). This completes the proof. ☐

5 Verifier geometry: logit margin, incoherence-driven max, and the loss-to-margin bridge

The phase-transition theorem of Section 8 compares the cumulative correct-logit signal against the worst-case incorrect-logit maximum. This section introduces the three analytic objects that underpin that comparison: the *logit margin* $M(x;Q)$ (Definition 5.1); a high-probability upper bound on the maximum incorrect-sequence logit, driven by martingale concentration of the trajectory plus the row-incoherence of W_U (Lemma 5.3, "Lemma A"); a high-probability lower bound on the maximum correct-sequence logit under the snowball assumption (Lemma 5.6, "Lemma B"); and a *loss-to-margin bridge* (Lemma 5.8) that converts the Lyapunov L-process analysis into a verifier-success statement on M. Throughout, $\|W_U^a\|_2$ denotes the Euclidean (ℓ_2) row norm of the sequence-embedding row of $a \in \mathcal{V}^n$ in the abstract sequence-unembedding matrix $W_U \in \mathbb{R}^{|\mathcal{V}|^n \times d}$ (cf. Remark 2.4).

5.1 The logit margin

Definition 5.1 (Logit margin)**.** For a question Q with answer set $\mathcal{A}(Q) \subseteq \mathcal{V}^n$ and a residual-stream state $x \in \mathbb{R}^d$, the *logit margin* is

$$M(x;Q) \;:=\; \max_{a \in \mathcal{A}(Q)} (W_U x)_a \;-\; \max_{a' \notin \mathcal{A}(Q)} (W_U x)_{a'}. \tag{13}$$

Remark 5.2 (Why the margin, not the loss, is the verifier's quantity)*.* The verifier consulted at inference time checks whether $\arg\max_{a \in \mathcal{V}^n} (W_U x)_a \in \mathcal{A}(Q)$, which holds iff $M(x;Q) > 0$. The constrained-softmax loss $L(x;Q) = -\log \pi(x;Q)$ of Definition 2.2 is the natural Lyapunov function for the gradient-flow analysis (its gradient has the clean form $\nabla L = W_U^\top (p - q_C)$ used implicitly throughout the proof of Theorem 8.1), but it does not coincide with verifier success: trajectories may have small L while still failing the argmax test if a single incorrect-answer mass is competitive. Lemma 5.8 below quantifies the relation in one direction (small loss implies positive margin), and the proof of Theorem 8.1 uses M as the success criterion to match the verifier's operational definition.

5.2 Lemma A: Gumbel max for incoherent logits

Lemma 5.3 (Max of incoherent logits along a random trajectory)**.** *Suppose Assumptions 3.4 and 3.9 hold with parameters $\mu_0 \in [0, 1/2]$, $R_U \geq 1$, and per-step value-norm bound M. Fix a horizon $T \geq 1$ and write the softmax-running-average trajectory as $x_T = \sum_{k=1}^{T} w_{T,k} V_k$ (Lemma 4.1), and let $\mathbb{E}[x_T]$ denote the expectation of x_T conditional on the question Q and the initial state (only the noise component of V_k is integrated out).* Assume the alignment hypothesis *of Eq.* (17) *(Lemma B below): on every effective step $E_t = 1$, the value vector V_t lies in $\mathrm{span}\{W_U^a : a \in \mathcal{A}\}$, so that $\mathbb{E}[x_T]$ is supported in the correct-sequence row span (cf. Remark 5.7). For any answer set $\mathcal{A} \subseteq \mathcal{V}^n$ with $1 \leq |\mathcal{A}| < |\mathcal{V}|^n$ and any $\delta \in (0,1)$:*

(i) Martingale-fluctuation bound (over the trajectory x_T)*. On an event of probability at least $1-\delta$ over the per-step noise in $(V_t)_{1 \leq t \leq T}$,*

$$\max_{a \notin \mathcal{A}} \langle W_U^a,\, x_T - \mathbb{E}[x_T] \rangle \;\leq\; 2\, R_U\, M \sqrt{\frac{T \,\log\!\big(2(|\mathcal{V}|^n - |\mathcal{A}|)/\delta\big)}{d}}, \tag{14}$$

where the radicand uses the loose quadratic-variation bound $\sum_t w_{T,t}^2 \leq 1 \leq T$ for compatibility with the horizon-scaling form used in the downstream Theorem 8.1; a tighter form replacing T by $\sum_t w_{T,t}^2$ in the radicand also holds (cf. Step 3 of the proof).

(ii) Drift bound (deterministic in W_U, via incoherence). *For every realisation of the trajectory,*

$$\max_{a \notin \mathcal{A}} \langle W_U^a, \mathbb{E}[x_T] \rangle \ \leq \ R_U \cdot \|\mathbb{E}[x_T]\|_2 \cdot \mu_0 \ \leq \ R_U \cdot M \cdot \mu_0, \tag{15}$$

where the first inequality uses Assumption 3.4 together with the alignment hypothesis (so that $\mathbb{E}[x_T]$ is aligned, up to row-norm rescaling, with the correct-sequence row span), and the second uses the deterministic norm bound $\|\mathbb{E}[x_T]\|_2 \leq M$ from the convex-combination representation: $\mathbb{E}[x_T] = \sum_k w_{T,k}\, \mathbb{E}[V_k]$ with $\sum_k w_{T,k} = 1$ and $\|\mathbb{E}[V_k]\|_2 \leq M$ gives, by convexity, $\|\mathbb{E}[x_T]\|_2 \leq \sum_k w_{T,k}\ \|\mathbb{E}[V_k]\|_2 \leq M$ (independent of T).

Combining (i) and (ii) by the triangle inequality $\langle W_U^a, x_T \rangle \leq \langle W_U^a, \mathbb{E}[x_T] \rangle + \langle W_U^a, x_T - \mathbb{E}[x_T] \rangle$, on the same event of probability $\geq 1 - \delta$,

$$\max_{a \notin \mathcal{A}} \langle W_U^a, x_T \rangle \ \leq \ R_U\, M \cdot \mu_0 \ + \ 2\, R_U\, M \sqrt{\frac{T\, \log\big(2(|\mathcal{V}|^n - |\mathcal{A}|)/\delta\big)}{d}}. \tag{16}$$

Proof. The argument has four stages: (S1) decompose x_T into drift plus martingale-difference; (S2) bound the drift's projection on any incorrect-sequence row via the incoherence μ_0; (S3) apply the Azuma-Hoeffding tail of Lemma 6.3 to the fluctuation, fixing one direction W_U^a at a time; (S4) discharge the union bound over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions.
Step 1: drift / fluctuation decomposition. Write $x_T = \mathbb{E}[x_T] + (x_T - \mathbb{E}[x_T])$. By linearity of the inner product, for every $a \in \mathcal{V}^n$,

$$\langle W_U^a, x_T \rangle \ = \ \langle W_U^a, \mathbb{E}[x_T] \rangle \ + \ \langle W_U^a, x_T - \mathbb{E}[x_T] \rangle\,.$$

The first summand is the cumulative *drift* from the effective component of the trajectory (the part of V_k aligned with $-\nabla L$ on effective steps); the second is a centred martingale-difference sum whose increments are $M_t^{(a)} := w_{T,t}\big(\langle W_U^a, V_t \rangle - \mathbb{E}[\langle W_U^a, V_t \rangle \mid \mathcal{F}_{t-1}]\big)$ with respect to the filtration $(\mathcal{F}_t)_{t \geq 0}$ of Section 2.6.
Step 2: drift bound via incoherence. By the alignment hypothesis in the lemma statement, the drift $\mathbb{E}[x_T]$ is supported in the span of the correct-sequence rows $\mathrm{span}\{W_U^b : b \in \mathcal{A}\}$ (cf. Remark 5.7: in the working stylisation Definition 2.6 the effective-step value vectors are proportional to $-\nabla L = W_U^\top(q_C - p)$, whose support lies in $\mathrm{span}\{W_U^b : b \in \mathcal{A}\}$, while noise-step contributions average to zero). For any $a \notin \mathcal{A}$ and any $b \in \mathcal{A}$, the row-incoherence bound of Definition 3.3 gives $\big|\langle W_U^a, W_U^b \rangle\big| \leq \mu_0\ \|W_U^a\|_2\ \|W_U^b\|_2 \leq \mu_0\, R_U^2$, and dividing by R_U to normalise one row gives $\big|\langle W_U^a / R_U, W_U^b \rangle\big| \leq \mu_0\, R_U$. Iterating over the convex combination defining $\mathbb{E}[x_T]$ and bounding by the trivial row-norm-times-incoherence factor:

$$\max_{a \notin \mathcal{A}} \langle W_U^a, \mathbb{E}[x_T] \rangle \ \leq \ R_U \cdot \|\mathbb{E}[x_T]\|_2 \cdot \mu_0,$$

where the explicit factor μ_0 is the maximum off-diagonal Gram entry between distinct rows (Definition 3.3). The tight norm bound $\|\mathbb{E}[x_T]\|_2 \leq M$ follows from the convex-combination representation $\mathbb{E}[x_T] = \sum_k w_{T,k}\, \mathbb{E}[V_k]$ together with $\|\mathbb{E}[V_k]\|_2 \leq \mathbb{E}\, \|V_k\|_2 \leq M$ (Assumption 3.9) and $\sum_k w_{T,k} = 1$: by convexity of the norm, $\|\mathbb{E}[x_T]\|_2 \leq \sum_k w_{T,k}\ \|\mathbb{E}[V_k]\|_2 \leq M \cdot \sum_k w_{T,k} = M$, *independent of* T. This proves Eq. (15).
Step 3: per-direction Azuma-Hoeffding fluctuation bound. Fix any $a \notin \mathcal{A}$. The sequence $(M_t^{(a)})_{1 \leq t \leq T}$ defined in Step 1 is a martingale-difference sequence with respect to $(\mathcal{F}_t)$. We verify its per-step bound and conditional variance:

- *Per-step boundedness.* By Cauchy-Schwarz and the triangle inequality, $\left|M_t^{(a)}\right| \le 2\, w_{T,t}\, \|W_U^a\|_2\, \|V_t\|_2 \le 2\, w_{T,t}\, R_U\, M$, using $\|W_U^a\|_2 \le R_U$ from Assumption 3.4 and $\|V_t\|_2 \le M$ from Assumption 3.9. Since $w_{T,t} \le 1$ under the running-average normalisation (Lemma 4.1), $\left|M_t^{(a)}\right| \le 2R_U M$ almost surely.
- *Conditional variance.* By Lemma 6.1 applied to the unit direction $e^{(a)} := W_U^a / \|W_U^a\|_2$ and the (conditionally mean-zero, isotropic-noise component of V_t under Definition 2.6; the effective component is absorbed into the drift), the conditional variance of $\langle W_U^a, V_t\rangle$ is at most $\|W_U^a\|_2^2 \cdot M^2/d \le R_U^2 M^2/d$. Hence $\mathbb{E}[(M_t^{(a)})^2 \mid \mathcal{F}_{t-1}] \le w_{T,t}^2 \cdot R_U^2 M^2/d$.

By Lemma 6.3 applied to the martingale-difference sequence $(M_t^{(a)})$ with $\sigma^2 = R_U^2 M^2$ (the proxy variance times d) and $c_M = 2R_U M$, together with $\sum_{t\le T} w_{T,t}^2 \le \sum_t w_{T,t} = 1 \le T$ (since $0 \le w_{T,t} \le 1$), we obtain that, for every per-direction failure budget $\delta' \in (0,1)$,

$$\mathbb{P}\left[\, |\langle W_U^a, x_T - \mathbb{E}[x_T]\rangle| \;>\; 2\, R_U\, M \sqrt{\frac{T\,\log(2/\delta')}{d}}\,\right] \;\le\; \delta'.$$

(The same bound holds with T replaced by $\sum_t w_{T,t}^2$ in the radicand, which is $\le T$; we keep T for compatibility with the rest of the paper.)

Step 4: explicit union bound over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions. We require simultaneous control on all $|\mathcal{V}|^n - |\mathcal{A}|$ events $\{\, |\langle W_U^a, x_T - \mathbb{E}[x_T]\rangle| > t\,\}$, one per $a \notin \mathcal{A}$. Set the per-direction failure budget to $\delta' := \delta/(|\mathcal{V}|^n - |\mathcal{A}|)$; then the per-direction bound of Step 3 reads

$$\mathbb{P}\left[\, |\langle W_U^a, x_T - \mathbb{E}[x_T]\rangle| \;>\; 2\, R_U\, M \sqrt{\frac{T\,\log\bigl(2(|\mathcal{V}|^n - |\mathcal{A}|)/\delta\bigr)}{d}}\,\right] \;\le\; \frac{\delta}{|\mathcal{V}|^n - |\mathcal{A}|}.$$

By the standard union bound applied to the family $\{\mathcal{E}^{(a)}\}_{a\notin\mathcal{A}}$ of failure events, indexed over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions,

$$\mathbb{P}\left[\, \exists\, a \notin \mathcal{A} :\; |\langle W_U^a, x_T - \mathbb{E}[x_T]\rangle| \;>\; 2\, R_U\, M \sqrt{\frac{T\,\log\bigl(2(|\mathcal{V}|^n - |\mathcal{A}|)/\delta\bigr)}{d}}\,\right] \;\le\; \sum_{a\notin\mathcal{A}} \frac{\delta}{|\mathcal{V}|^n - |\mathcal{A}|} \;=\; \delta.$$

This is exactly Eq. (14). Combining with the deterministic drift bound of Step 2 (Eq. (15)) and the triangle inequality $\langle W_U^a, x_T\rangle \le \langle W_U^a, \mathbb{E}[x_T]\rangle + \langle W_U^a, x_T - \mathbb{E}[x_T]\rangle$ yields Eq. (16) on the same event of $\mathbb{P} \ge 1 - \delta$. This completes the proof. □

Remark 5.4 (Where the $\sqrt{\log(|\mathcal{V}|^n)/d}$ scaling comes from). The $\sqrt{\log(|\mathcal{V}|^n - |\mathcal{A}|)/d}$ scaling in Eq. (14) arises from two sources, both *probabilistic* in the trajectory x_T (not deterministic in W_U): the $1/d$ factor is the high-dimensional orthogonality variance of Lemma 6.1 (each noise step contributes variance $\le M^2/d$ to a fixed projection $\langle W_U^a, V_t\rangle$ after row-norm rescaling), and the $\log(|\mathcal{V}|^n - |\mathcal{A}|)/\delta$ factor is the union-bound budget of Step 4 distributed across the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions. The incoherence μ_0 enters only in Eq. (15) as a deterministic contribution from the drift's projection onto an incorrect-sequence row, not as a noise-floor correction.

Remark 5.5 (Probabilistic in the trajectory; deterministic in W_U). Lemma 5.3 treats W_U as a deterministic object characterised by the two bounds in Assumption 3.4 (incoherence $\le \mu_0$ and row norms $\le R_U$); no random-matrix postulate on W_U is used. The probability statement is over the per-step noise in $(V_t)_{1\le t\le T}$, i.e. the random reasoning trajectory. For the abstract sequence-unembedding linearisation of Remark 2.4 with empirically measurable $\mu(W_U)$, the bound is directly applicable; the $\mu_0 = 1/2$ worst-case bound in Assumption 3.4 is loose by typically a factor $O(\sqrt{n \log |\mathcal{V}|/d})$ in practice (Remark 3.5).

5.3 Lemma B: signal accumulation under the snowball assumption

The complement of Lemma 5.3 is a lower bound on the maximum correct-sequence logit. The (SS) snowball assumption supplies the per-step rate of effective tokens; on each effective step the value vector V_j is by hypothesis aligned with $-\nabla L(x_{j-1}; Q)$ in the working stylisation of Definition 2.6, and we record the consequence for the correct-sequence logits in this lemma.

Lemma 5.6 (Signal accumulation)**.** *Suppose Assumptions 3.1, 3.4 and 3.9 hold, and assume the working alignment condition: there is an absolute constant* $\cos\theta_0 \in (0, 1]$ *such that on every effective step* $E_t = 1$ *(in the sense of Definition 2.6), the value vector* V_t *satisfies*

$$\max_{a \in \mathcal{A}(Q)} \langle W_U^a, V_t \rangle \;\geq\; \cos\theta_0 \cdot \|V_t\|_2 \cdot \max_{a \in \mathcal{A}(Q)} \|W_U^a\|_2 \,, \tag{17}$$

i.e. V_t *has positive component along at least one correct-sequence row of* W_U*. Fix any horizon* $T \geq 1$ *and* $\delta \in (0, 1)$*. Define the* average per-step softmax weight

$$\tau \;:=\; T^{-1} \sum_{t=1}^{T} w_{T,t} \;=\; \frac{1}{T}, \tag{18}$$

where the second equality uses $\sum_t w_{T,t} = 1$ *from Definition 2.1; in particular* $\tau = 1/T$ *identically,* independent of the specific softmax weight distribution*, since the running-average representation Eq.* (12) *forces the weights to be a probability vector. Then on an event of* $\mathbb{P} \geq 1 - \delta$ *over the per-step noise:*

$$\max_{a \in \mathcal{A}(Q)} \langle W_U^a, x_T \rangle \;\geq\; \lambda_0(Q) \cdot \cos\theta_0 \;-\; 2\, R_U\, M \sqrt{\log(2/\delta)/d}, \tag{19}$$

where the signal $\lambda_0(Q) \cdot \cos\theta_0 = \lambda_0(Q) \cdot T\tau \cdot \cos\theta_0$ *is the running-weighted effective contribution (the* $T\tau = 1$ *factor reflects the convex-combination structure of the trajectory), and the noise bound uses the row-norm bound* $\left\|W_U^{a^\star}\right\|_2 \leq R_U$ *together with the conservative martingale-variance bound* $\sum_t w_{T,t}^2 \leq \sum_t w_{T,t} = 1$*.*

Proof. The proof splits into the signal mean (effective-step contribution) and the noise-step concentration. Throughout, $T\tau = \sum_t w_{T,t} = 1$ identically (Eq. (18)); we keep both notations side-by-side to highlight the convex-combination structure.

Step 1: signal from effective steps. By Lemma 4.1, the residual stream after T steps is $x_T = \sum_{k=1}^{T} w_{T,k}\, V_k$. Fixing any $a^\star \in \arg\max_{a \in \mathcal{A}(Q)} \|W_U^a\|_2$, the correct-sequence logit at step T is

$$\left\langle W_U^{a^\star}, x_T \right\rangle \;=\; \sum_{k=1}^{T} w_{T,k} \left\langle W_U^{a^\star}, V_k \right\rangle.$$

Decompose the sum by the effective-token indicator E_k:

$$\left\langle W_U^{a^\star}, x_T \right\rangle \;=\; \sum_{k: E_k = 1} w_{T,k} \left\langle W_U^{a^\star}, V_k \right\rangle \;+\; \sum_{k: E_k = 0} w_{T,k} \left\langle W_U^{a^\star}, V_k \right\rangle.$$

On each effective step, by hypothesis Eq. (17) applied with $a = a^\star$, $\left\langle W_U^{a^\star}, V_k \right\rangle \geq \cos\theta_0 \, \|V_k\|_2 \left\| W_U^{a^\star} \right\|_2 \geq \cos\theta_0$, where the last inequality used the row-norm bound $\left\| W_U^{a^\star} \right\|_2 \geq 1$ which follows from Assumption 3.4 (the row norms lie in $[1, R_U]$ in the convention adopted there). Hence

$$\sum_{k:E_k=1} w_{T,k} \left\langle W_U^{a^\star}, V_k \right\rangle \;\geq\; \cos\theta_0 \cdot \sum_{k:E_k=1} w_{T,k}.$$

In the snowball region, the (SS) assumption ensures $\mathbb{E}[E_k \mid \mathcal{F}_{k-1}] \geq \lambda_0$ for k with $L(x_{k-1}; Q) < L^\star(Q)$, so by linearity of expectation,

$$\mathbb{E}\Bigg[\sum_{k \leq T} w_{T,k} E_k\Bigg] \;\geq\; \lambda_0 \cdot \sum_{k \leq T} w_{T,k} \;=\; \lambda_0 \cdot T\tau \;=\; \lambda_0,$$

where the last equality uses $T\tau = 1$. This is the deterministic part of Eq. (19): the expected signal $\lambda_0 \cdot \cos\theta_0$ is *independent of* T (as expected for a convex-combination trajectory whose weights renormalise to 1 irrespective of horizon).

Step 2: noise-step concentration. On noise steps ($E_k = 0$), the value vector V_k has conditional distribution that is symmetric in the sense of Definition 2.6 (the noise component is mean-zero conditional on $\mathcal{F}_{k-1}$ in the high-dimensional model). The quantity

$$D_T \;:=\; \sum_{k=1}^{T} w_{T,k} \left(\left\langle W_U^{a^\star}, V_k \right\rangle - \mathbb{E}\left[\left\langle W_U^{a^\star}, V_k \right\rangle \mid \mathcal{F}_{k-1}\right] \right),$$

is a martingale-difference sum (with respect to $\mathcal{F}_T$) and satisfies the per-step bound

$$\left| w_{T,k} \cdot \left(\left\langle W_U^{a^\star}, V_k \right\rangle - \mathbb{E}[\cdot] \right) \right| \;\leq\; 2\, w_{T,k} \left\| W_U^{a^\star} \right\|_2 \|V_k\|_2 \;\leq\; 2\, w_{T,k}\, R_U\, M,$$

by Assumption 3.9 and the row-norm bound. The conditional-variance bound on each term follows from Lemma 6.1: each centred term has variance at most $4 w_{T,k}^2 \left\| W_U^{a^\star} \right\|_2^2 \|V_k\|_2^2 / d \leq 4 w_{T,k}^2 R_U^2 M^2 / d$, from the high-d orthogonality of the noise direction to the fixed row $W_U^{a^\star}$. Hence the conditional quadratic variation

$$\sum_{k \leq T} \mathbb{E}[D_k^2 \mid \mathcal{F}_{k-1}] \;\leq\; \frac{4 R_U^2 M^2}{d} \cdot \sum_{k \leq T} w_{T,k}^2 \;\leq\; \frac{4 R_U^2 M^2}{d},$$

where the final bound uses $\sum_k w_{T,k}^2 \leq \sum_k w_{T,k} = 1$ (since $0 \leq w_{T,k} \leq 1$ for every k). Applying the Bernstein-for-martingales bound [5] with this conservative quadratic-variation estimate, for any $\delta \in (0, 1)$,

$$\mathbb{P}\Big[\, |D_T| > 2 R_U M \sqrt{\log(2/\delta)/d} \,\Big] \;\leq\; \delta.$$

Step 3: union-bound discharge. The $1 - \delta$ probability in the lemma statement is discharged by a single application of Bernstein's martingale tail bound to the scalar martingale D_T. There is exactly one event whose failure budget is consumed: $\mathcal{E}_{\text{noise}} := \{ |D_T| \leq 2 R_U M \sqrt{\log(2/\delta)/d} \}$, which is the Bernstein-bound output of Step 2. Hence by the union bound over the singleton family $\{\mathcal{E}_{\text{noise}}\}$, $\mathbb{P}[\mathcal{E}_{\text{noise}}] \geq 1 - \delta$. No further events are introduced in the assembly, so the $1 - \delta$ inherited from $\mathcal{E}_{\text{noise}}$ is the unique high-probability budget of the lemma.

Step 4: assemble. On the event $\mathcal{E}_{\text{noise}}$ of $\mathbb{P} \geq 1 - \delta$, combining Steps 1 and 2 yields

$$\left\langle W_U^{a^\star}, x_T \right\rangle \;\geq\; \cos\theta_0 \cdot \lambda_0 \;-\; 2 R_U M \sqrt{\log(2/\delta)/d},$$

which is Eq. (19) since $\max_{a \in \mathcal{A}(Q)} \langle W_U^a, x_T \rangle \geq \langle W_U^{a^\star}, x_T \rangle$. This completes the proof. □

Remark 5.7 (The alignment condition Eq. (17)). The hypothesis Eq. (17) is the working stylisation of the effective-step mode: each effective-step value vector has positive cosine with at least one correct-sequence unembedding row, and (the stronger consequence used in Lemma A's drift bound) $V_t \in \mathrm{span}\{W_U^a : a \in \mathcal{A}\}$ on every effective step. This is a structural consequence of Definition 2.6 in the model where $V_t = -\nabla L(x_{t-1}; Q)/\|\nabla L(x_{t-1}; Q)\|_2$, since $-\nabla L = W_U^\top(q_C - p)$ is a linear combination of sequence-unembedding rows with non-zero components only on $\mathcal{A}(Q)$ (the conditional posterior q_C is supported on $\mathcal{A}(Q)$, so the residual $q_C - p$ contributes a non-zero coefficient only on correct-sequence rows after the $W_U^\top$ pull-back). The constant $\cos\theta_0$ is the worst-case cosine of V_t with the closest correct row over the trajectory; we treat it as a fixed model-dependent constant in the framework. Both Lemmas A and B invoke this hypothesis: Lemma B for the lower bound on $\langle W_U^{a^\star}, V_t\rangle$ on effective steps, and Lemma A for the support property of $\mathbb{E}[x_T]$ that enables the incoherence-based drift bound.

5.4 Loss-to-margin bridge

The Lyapunov supermartingale analysis of Section 8 controls the loss process $L(x_t; Q)$. The verifier checks margin positivity $M(x_t; Q) > 0$. The next lemma bridges the two: when the loss drops below an explicit decode threshold $L_{\mathrm{dec}}(|\mathcal{V}|^n, |\mathcal{A}|)$, the margin is positive.

Lemma 5.8 (Loss-to-margin bridge). *For every $x \in \mathbb{R}^d$ and every question Q with $|\mathcal{A}(Q)| \in \{1, \ldots, |\mathcal{V}|^n - 1\}$, if*

$$L(x; Q) \; < \; L_{\mathrm{dec}}(|\mathcal{V}|^n, |\mathcal{A}(Q)|), \qquad L_{\mathrm{dec}}(|\mathcal{V}|^n, k) \; := \; \log\Big(1 + \tfrac{1}{k}\Big), \tag{20}$$

then $M(x; Q) > 0$. In particular, for $|\mathcal{A}(Q)| = 1$, $L_{\mathrm{dec}} = \log 2$.

Proof. The proof is a two-line pigeonhole on the softmax probabilities over the effective alphabet $\mathcal{V}^n$.

Step 1: loss bound implies correct-mass bound. By Definition 2.2, $L(x; Q) = -\log \pi(x; Q)$ where $\pi(x; Q) = \sum_{a \in \mathcal{A}(Q)} \mathrm{softmax}(W_U x)_a$. The hypothesis $L(x; Q) < \log(1 + 1/k)$ with $k = |\mathcal{A}(Q)|$ rearranges to $\pi(x; Q) > 1/(1 + 1/k) = k/(k+1)$.

Step 2: pigeonhole between correct and incorrect masses. The total softmax mass over $\mathcal{V}^n$ is 1, so the incorrect-sequence mass is $1 - \pi(x; Q) < 1 - k/(k+1) = 1/(k+1)$. By an averaging argument,

$$\max_{a \in \mathcal{A}(Q)} \mathrm{softmax}(W_U x)_a \; \geq \; \frac{\pi(x; Q)}{|\mathcal{A}(Q)|} \; > \; \frac{k/(k+1)}{k} \; = \; \frac{1}{k+1},$$

while $\max_{a' \notin \mathcal{A}(Q)} \mathrm{softmax}(W_U x)_{a'} \leq 1 - \pi(x; Q) < 1/(k+1)$. Therefore

$$\max_{a \in \mathcal{A}} \mathrm{softmax}(W_U x)_a \; > \; \max_{a' \notin \mathcal{A}} \mathrm{softmax}(W_U x)_{a'}.$$

Since softmax is a strictly monotone transform of the logits (it preserves order), the same inequality holds for the logits themselves: $\max_{a \in \mathcal{A}} (W_U x)_a > \max_{a' \notin \mathcal{A}} (W_U x)_{a'}$, which by Definition 5.1 is $M(x; Q) > 0$. This completes the proof. □

Remark 5.9 (Why L_{dec} shrinks with $|\mathcal{A}|$). The threshold $L_{\mathrm{dec}}(|\mathcal{V}|^n, k) = \log(1 + 1/k)$ decreases with $|\mathcal{A}| = k$: for a singleton correct set ($k = 1$), $L_{\mathrm{dec}} = \log 2 \approx 0.693$; for $k = 10$ correct sequences, $L_{\mathrm{dec}} \approx 0.0953$; for $k = 100$, $L_{\mathrm{dec}} \approx 0.00995$. Intuitively, a larger correct set is "easier to hit" in mass, but harder to dominate per-sequence: each individual correct mass is at most $1/k$, so the margin inequality requires a tighter loss bound. The explicit form $\log(1 + 1/k)$ is the optimal threshold for the averaging bound in Step 2 of the proof; it is unimprovable without further information on the distribution of mass within $\mathcal{A}(Q)$ (cf. Remark 11.6).

Remark 5.10 (Direction of the bridge). Lemma 5.8 states the implication $L < L_{\text{dec}} \Rightarrow M > 0$. The reverse implication ($M > 0 \Rightarrow L <$ some threshold) is false: a trajectory can have $M > 0$ while L is moderate (since $M > 0$ only requires the max correct-sequence logit to slightly exceed the max incorrect-sequence logit, but the loss penalises the entire distribution). The proof of Theorem 8.1 uses only the forward direction: the snowball-branch analysis drives L below L_{dec}, then Lemma 5.8 delivers verifier success. This one-sided bridge is the cleanest separation of "Lyapunov dynamics" (the L analysis) from "verifier success" (the M statement) in the framework.

6 High-dimensional concentration of the radial walk

The phase-transition argument of Section 8 requires high-probability control on cumulative radial fluctuations over the trajectory horizon. This section assembles two facts: (i) the $\Theta(1/d)$ projection variance of the per-step noise onto a fixed direction (Lemma 6.1), and (ii) the cumulative martingale-tail bound on radial fluctuations (Lemma 6.3). Together these provide the union-bound control needed to discharge the $1 - \delta$ in Theorem 8.1.

6.1 High-dimensional orthogonality

Lemma 6.1 (Projection variance and concentration on the sphere)**.** *Let $u \in \mathbb{R}^d$ be uniformly distributed on the sphere S^{d-1} and let $e \in \mathbb{R}^d$ be any deterministic unit vector. Then*

$$\mathbb{E}\,\langle e, u\rangle^2 \;=\; \frac{1}{d}, \tag{21}$$

and for every $t \in [0, c'\sqrt{d}\,]$ with $c' \in (0, 1)$ an absolute constant,

$$\mathbb{P}\Big[\; |\langle e, u\rangle| \;>\; \frac{t}{\sqrt{d}} \;\Big] \;\le\; 2\exp(-c\,t^2), \tag{22}$$

for an absolute constant $c > 0$.

Proof of Lemma 6.1, sketch with citation. Eq. (21) follows from the rotational invariance of the uniform measure: $\mathbb{E}[uu^\top] = (1/d)I$, so $\mathbb{E}\,\langle e, u\rangle^2 = e^\top\mathbb{E}[uu^\top]e = \|e\|^2/d = 1/d$. Eq. (22) is the standard sub-Gaussian-on-sphere concentration: by Theorem 3.4.6 of [12] (Section 3.4, "Concentration on the sphere"), the random variable $\langle e, u\rangle$ is sub-Gaussian with proxy variance C/d, hence satisfies the stated tail bound. This completes the proof. □

Remark 6.2 (The structural role of $1/d$). Eq. (21) is the precise quantitative form of the slogan "in high dimensions, random vectors are nearly orthogonal". It is the foundation of the noise-variance bound $\mathbb{E}[M_t^2 \mid \mathcal{F}_{t-1}] \le \sigma^2/d$ used in the cumulative-noise concentration bound of Lemma 6.3 below. The scaling-limit framework of [1] (Section 2.2) identifies precisely this $1/d$ factor as the source of the diffusion coefficient in the $d \to \infty$ SDE limit of online SGD trajectories.

6.2 Cumulative radial-walk concentration

Lemma 6.3 (Azuma-Hoeffding tail for cumulative radial fluctuations)**.** *Let $(M_t)_{t\ge1}$ be a martingale-difference sequence with respect to a filtration $(\mathcal{F}_t)_{t\ge0}$ satisfying $\mathbb{E}[M_t^2 \mid \mathcal{F}_{t-1}] \le 2\sigma^2/d$ and $|M_t| \le c_M$*

almost surely for a uniform constant $c_M \le 2\sigma$ (in this paper's applications, M_t is the centred projection of the per-step value vector V_t on a fixed unit direction; the variance bound follows from Lemma 6.1 and the almost-sure bound from Assumption 3.9). Then for every horizon $T \ge 1$ and every $\delta \in (0,1)$,

$$\mathbb{P}\left[\max_{1\le t\le T}\left|\sum_{s=1}^{t} M_s\right| \;>\; 2\sigma\sqrt{\frac{T\,\log(2/\delta)}{d}}\;\right] \;\le\; \delta. \tag{23}$$

Proof. The argument is the standard application of the Azuma-Hoeffding martingale tail bound, with two stages: a maximal-inequality reduction and the exponential-supermartingale calculation.
Step 1: maximal inequality. By Doob's maximal inequality applied to the non-negative submartingale $\exp(\lambda\sum_{s\le t} M_s)$ (for any $\lambda > 0$),

$$\mathbb{P}\left[\max_{t\le T}\sum_{s\le t} M_s \;\ge\; u\right] \;=\; \mathbb{P}\left[\max_{t\le T}\exp\Big(\lambda\sum_{s\le t} M_s\Big) \;\ge\; e^{\lambda u}\right] \;\le\; e^{-\lambda u}\,\mathbb{E}\left[\exp\Big(\lambda\sum_{s\le T} M_s\Big)\right].$$

Step 2: exponential MGF bound. By the sub-Gaussian Azuma-Hoeffding lemma applied to the bounded-difference martingale $(\sum_{s\le t} M_s)_{t\ge 0}$ with differences $|M_t| \le c_M$,

$$\mathbb{E}\left[\exp\Big(\lambda\sum_{s\le T} M_s\Big)\right] \;\le\; \exp\Big(\frac{\lambda^2 T c_M^2}{2}\Big).$$

A sharper bound replacing c_M^2 with the conditional variance $\mathbb{E}[M_t^2 \mid \mathcal{F}_{t-1}] \le 2\sigma^2/d$ gives $\mathbb{E}\exp(\lambda\sum M_s) \le \exp(\lambda^2 T\cdot 2\sigma^2/d/2) = \exp(\lambda^2 T\sigma^2/d)$; this is the Bernstein-form refinement provided by [5].
Step 3: optimise λ. Combining Steps 1 and 2, $\mathbb{P}[\max_{t\le T}\sum_{s\le t} M_s \ge u] \le \exp(-\lambda u + \lambda^2 T\sigma^2/d)$. The right-hand side is minimised at $\lambda = ud/(2T\sigma^2)$, yielding

$$\mathbb{P}\left[\max_{t\le T}\sum_{s\le t} M_s \;\ge\; u\right] \;\le\; \exp\Big(-\frac{u^2 d}{4T\sigma^2}\Big).$$

A two-sided union bound (applying the same argument to $-\sum M_s$) doubles the tail probability, yielding $\mathbb{P}[\max_{t\le T}\left|\sum_{s\le t} M_s\right| \ge u] \le 2\exp(-u^2 d/(4T\sigma^2))$. Setting the right-hand side to δ gives $u^2 d/(4T\sigma^2) = \log(2/\delta)$, hence $u = 2\sigma\sqrt{T\log(2/\delta)/d}$, which is Eq. (23). This completes the proof.
Constant tracking (post-2026-05-26 audit). An earlier version of this lemma stated the bound as $\sigma\sqrt{2T\log(2/\delta)/d}$, off from the correct $2\sigma\sqrt{T\log(2/\delta)/d}$ by a factor of $\sqrt{2}$. The corrected constant $2\sigma\sqrt{T\log(2/\delta)/d}$ is the form used throughout the subsequent theorem statements; downstream bounds in Section 8 have been adjusted accordingly. □

Remark 6.4 (Sharp $1/\sqrt{d}$ scaling of fluctuations). Eq. (23) gives the cumulative fluctuation magnitude as $\Theta\big(\sigma\sqrt{T\log(1/\delta)/d}\big)$ in horizon T and dimension d. This $1/\sqrt{d}$ scaling is the quantitative manifestation of the high-dimensional orthogonality of Lemma 6.1, and it is the budget that gets compared against the cumulative inward drift in the proof of Theorem 8.1 (snowball branch). The comparison is made precise in Section 8; the consequence is the headline $1/\sqrt{d}$ scaling of the critical effective-token rate.

7 Galton-Watson coupling for the extinction branch

The extinction branch of Theorem 8.1 (sub-critical $\lambda_0 < \lambda_c$) is proved by coupling the effective-token chain on the constrained-softmax loss process to a Galton-Watson branching process and applying the classical sub-critical trichotomy. This section is the v3 rewrite of the coupling: the chain runs on the loss $L_t := L(x_t; Q)$ (rather than on the radial coordinate as in v2), the offspring mean is the explicit ratio λ_0/λ_c derived in Section 8, and the extinction statement bounds the probability that the verifier-success event $\mathcal{S} := \{\exists T \leq T_{\max} : M(x_T; Q) > 0\}$ occurs in the sub-critical regime.

7.1 Galton-Watson process on the loss-effective chain

Definition 7.1 (Comparison Galton-Watson process)**.** Fix $d \geq 1$, $|\mathcal{V}| \geq 2$, $|\mathcal{A}| \in \{1, \ldots, |\mathcal{V}|^n - 1\}$, horizon $T_{\max} \geq 1$, and $\lambda_0 \in (0, 1]$. Let $\lambda_c := \lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})$ denote the critical-rate functional of Theorem 8.1 below and define the offspring-mean parameter

$$m \;:=\; \frac{\lambda_0}{\lambda_c} \;\in\; (0, \infty).$$

The *comparison Galton-Watson process* $(Z_n)_{n\geq 0}$ is the discrete-time branching process with $Z_0 = 1$ and offspring distribution Poisson(m) at every node.

Remark 7.2 (Why Poisson). The Poisson choice is the canonical mean-m offspring distribution under which the per-generation sub/super-critical trichotomy reduces to the closed-form generating-function identity below. The coupling of Lemma 7.4 bounds the effective-chain generation size by stochastic domination, and the Poisson form is the target of the standard Bernoulli-to-Poisson law-of-small-numbers domination. The structural conclusions (extinction certain for $m < 1$, expected generation size m^n) carry over to any offspring distribution with mean m and bounded variance; see [12] Ch. 5 for a textbook treatment of the Galton-Watson trichotomy.

Fact 7.3 (Sub-critical Galton-Watson trichotomy)**.** For the process of Definition 7.1 with $m < 1$:

(i) *Extinction certain.* $\mathbb{P}[\exists n : Z_n = 0] = 1$.

(ii) *Expected size decays geometrically.* $\mathbb{E}Z_n = m^n$ for every $n \geq 0$; in particular, $\sum_{n\geq 0} \mathbb{E}Z_n = 1/(1-m) < \infty$.

(iii) *Tail of total progeny.* The total progeny $\mathsf{Z} := \sum_{n\geq 0} Z_n$ satisfies $\mathbb{E}\mathsf{Z} = 1/(1-m)$ and $\mathbb{P}[\mathsf{Z} > N] \leq m^N$ for every $N \geq 1$.

Fact 7.3 is standard textbook material; (i)-(ii) appear in Athreya-Ney, *Branching Processes*, Chapter 1, Theorem 1, and (iii) follows from generating-function manipulation (Athreya-Ney Chapter 1, Theorem 3) and the elementary identity $m^N = \sum_{n=N}^{\infty} m^n(1-m)$.

7.2 Coupling lemma

Lemma 7.4 (Sub-critical extinction via Galton-Watson coupling)**.** *Suppose Assumptions 3.1, 3.4, 3.7 and 3.9 hold. Fix $d \geq 1$, $|\mathcal{V}| \geq 2$, $T_{\max} \geq 1$, and $\delta_- \in (0, 1)$. If*

$$\lambda_0 \;<\; \lambda_c(d, |\mathcal{V}|, |\mathcal{A}(Q)|, T_{\max}), \tag{24}$$

then there exists a coupling of the effective-token chain $(E_t)_{1\leq t\leq T_{\max}}$ (Definition 2.6) to a sub-critical Galton-Watson process $(Z_n)_{n\geq 0}$ with offspring mean $m = \lambda_0/\lambda_c < 1$ such that the total number of *effective tokens emitted before horizon $T_{\max}$ is stochastically dominated by the total progeny Z of (Z_n). Consequently,*

$$\mathbb{P}\big[\,\mathcal{S}\,\big] \;=\; \mathbb{P}\big[\,\exists\, T \le T_{\max} : M(x_T; Q) > 0\,\big] \;\le\; \delta_-(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}, \lambda_0), \tag{25}$$

where the explicit extinction bound is

$$\delta_-(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}, \lambda_0) \;\coloneqq\; m^{N(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})} \;+\; \delta_{\text{noise}}(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}),$$

with $N(\cdot)$ the minimum number of effective tokens required to drive the loss below $L_{\text{dec}}(|\mathcal{V}|^n, |\mathcal{A}|)$ (Lemma 5.8) and δ_{noise} the failure probability of the noise-control event in Lemma 6.3.

Proof. The proof has three stages: (a) define the generation partition, (b) dominate generation-n size by Poisson(m), and (c) translate sub-critical extinction into a verifier-success-failure statement via the loss-to-margin bridge.

Stage 1: generation partition. Partition the time axis $\{1, 2, \ldots, T_{\max}\}$ into *generation windows* of length $\Delta \coloneqq T_{\max}/N$ where $N \coloneqq N(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})$ is the minimum number of effective tokens required to reduce the loss from $L_0 \le L^\star(Q)$ to $L_{\text{dec}}(|\mathcal{V}|^n, |\mathcal{A}|)$ under the snowball contraction rate (computed explicitly in the proof of Theorem 8.1, snowball branch). The generation boundaries are deterministic, and each generation contains exactly Δ time steps. Generation 0 is empty: it represents the "seed" effective token, which is the first $t \ge 1$ with $E_t = 1$ (this exists almost surely under Assumption 3.1 with $\lambda_0 > 0$; if no such t exists by time $T_{\max}$, then $\mathcal{S}$ trivially fails and the lemma's conclusion holds vacuously).

Stage 2: Poisson domination per generation. By Assumption 3.1, the conditional distribution of generation-n effective-token count $Z_n^{\text{eff}} \coloneqq \sum_{t \in \text{gen}_n} E_t$ given the filtration at the start of generation n is a sum of Δ Bernoulli indicators each with conditional success probability $\lambda(L_t) \le \lambda_0$ on the snowball region (the upper-bound side of Assumption 3.1). Hence

$$\mathbb{E}[Z_n^{\text{eff}} \mid \mathcal{F}_{(n-1)\Delta}] \;\le\; \Delta \cdot \lambda_0 \;=\; \frac{T_{\max}}{N} \cdot \lambda_0 \;=\; \frac{\lambda_0}{\lambda_{\text{c}}} \;=\; m,$$

where the third equality used $N = T_{\max} \cdot \lambda_{\text{c}}$ (definitional, by construction of N as the snowball-required count; see Step 3 below for the verification of this identity from the snowball-branch proof). By the classical Bernoulli-to-Poisson stochastic domination (Lindvall, *Lectures on the Coupling Method*, Theorem 3.7), a sum of Δ independent Bernoullis with parameters p_i summing to at most m is stochastically dominated by Poisson(m). Chaining across generations via the strong Markov property at each generation boundary, the entire chain $(Z_n^{\text{eff}})_{n \ge 1}$ is stochastically dominated by the Galton-Watson process $(Z_n)_{n \ge 0}$ of Definition 7.1.

Stage 3: invoke trichotomy and translate to $\mathcal{S}$. By Fact 7.3(iii), the total progeny Z of the GW process satisfies $\mathbb{P}[\mathsf{Z} \ge N] \le m^N$. By the stochastic-domination established in Stage 2, the total number of effective tokens emitted by time $T_{\max}$ is $\mathsf{Z}^{\text{eff}} \coloneqq \sum_{n \ge 1} Z_n^{\text{eff}} \preceq \mathsf{Z}$, hence $\mathbb{P}[\mathsf{Z}^{\text{eff}} \ge N] \le m^N$. By Step 1's definition of N, the event $\{\mathsf{Z}^{\text{eff}} < N\}$ is exactly the event that the trajectory has not accumulated enough effective tokens to reduce L below $L_{\text{dec}}(|\mathcal{V}|^n, |\mathcal{A}|)$ along the snowball-contraction trajectory; on this event, the loss-to-margin bridge (Lemma 5.8) is *not* triggered, and therefore $M(x_t; Q) > 0$ is not guaranteed at any $t \le T_{\max}$.

A complication: $\mathsf{Z}^{\text{eff}} < N$ alone does not imply $M(x_t; Q) \le 0$ uniformly; the trajectory may achieve margin positivity via a favourable noise fluctuation. To control this, we require the noise-fluctuation event of Lemma 6.3. By the displayed bound in that lemma applied at horizon $T_{\max}$ and probability δ_{noise}, on an event of $\mathbb{P} \ge 1 - \delta_{\text{noise}}$, the cumulative noise contribution to $\langle W_U^a, x_t \rangle$ for any $a \in \mathcal{A}(Q)$ is bounded by $2\,\|W_U\|_{\text{op}} \sqrt{T_{\max} \log(2/\delta_{\text{noise}})/d}$, which is by hypothesis (Eq. (24)

and the explicit form of λ_c from Section 8) smaller in magnitude than the cumulative incorrect-logit fluctuation. Combining $\{\mathsf{Z}^{\mathrm{eff}} < N\}$ with the noise-control event by a union bound (over the two events) gives $\mathbb{P}[\mathcal{S}] \le m^N + \delta_{\mathrm{noise}}$, which is Eq. (25) with the stated δ_-. □

Remark 7.5 (Behaviour at the critical point). At exact criticality $\lambda_0 = \lambda_c$ (offspring mean $m = 1$), the GW process is critical: extinction is certain but the expected total progeny is infinite. The coupling of Lemma 7.4 therefore gives only a one-sided statement at exact criticality, and Theorem 8.1 is stated for *strict* sub-criticality $\lambda_0 < \lambda_c \cdot (1-\eta)$ with slack $\eta > 0$ to avoid invoking the critical fragility window. The empirical validation sweep at $|\mathcal{V}| = 256$ accompanying this paper (Remark 11.3) measures a window of width $\Theta(1/\sqrt{T_{\max} \log |\mathcal{V}|})$ around criticality, consistent with the GW critical-fluctuation scaling.

Remark 7.6 (Stochastic domination vs. exact coupling). The coupling in Lemma 7.4 is one-sided: the generation-size sequence Z_n^{eff} is dominated by, not equal to, the GW process. This is sufficient for the sub-critical-extinction conclusion (which is monotone in offspring size): if the dominating process goes extinct, so does the dominated one. The dominance is strict in the sense that the per-step effective rate $\lambda(L_t)$ may be strictly less than λ_0 when $L_t < L_{t-1}$ (the snowball region is exited downward, reducing λ), but this only makes the extinction conclusion stronger.

8 Theorem T1: sharp phase transition in verifier-margin success

This section states and proves the headline result of the framework: a sharp phase transition in the verifier-success probability at the critical effective-token rate $\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) = c_1 \sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}\, d)}$, where c_1 is an explicit absolute constant arising from the product of the martingale-concentration constants in Lemma 5.3 (Lemma A; its per-direction Azuma-Hoeffding bound plus union bound over $|\mathcal{V}|^n - |\mathcal{A}|$ directions) and the martingale-tail constants in Lemma 6.3. The proof has two branches: (i) a snowball branch (super-critical, $\lambda_0 > \lambda_c$) combining the signal-accumulation bound (Lemma 5.6) with the incoherent-logit max bound (Lemma 5.3) and the loss-to-margin bridge (Lemma 5.8); (ii) an extinction branch (sub-critical, $\lambda_0 < \lambda_c$) via the Galton-Watson coupling of Lemma 7.4.

8.1 Statement

Theorem 8.1 (Sharp phase transition in verifier success)**.** *Suppose Assumptions 3.1, 3.4, 3.7 and 3.9 hold. Fix $d \ge 1$, $|\mathcal{V}| \ge 2$, a question Q with $|\mathcal{A}(Q)| \in \{1, \ldots, |\mathcal{V}|^n - 1\}$ and $\lambda_0(Q) > 0$, horizon $T_{\max} \ge 1$, and $\delta \in (0, 1)$. Define the critical effective-token rate*

$$\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) \;:=\; c_1 \cdot \sqrt{\frac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{T_{\max} \cdot d}}, \tag{26}$$

where

$$c_1 \;:=\; \frac{2\sqrt{2}\, R_U\, M\, \kappa(\mu_0)}{\cos\theta_0}, \tag{27}$$

is the explicit absolute constant assembled from the martingale-concentration constants of Lemmas 5.3 and 5.6, with $\kappa(\mu_0) := 1 + \mu_0$ absorbing the incoherence-drift contribution as a $O(1)$ multiplicative correction on the leading noise term (under the regime $T_{\max} \ge T_0 := \log(2/\delta)$ stated in part (i)), and $\cos\theta_0$ the alignment-cosine of Eq. (17). The constant c_1 has the form $c_1 = (\text{numerical constant}) \cdot R_U M \kappa(\mu_0)/\cos\theta_0$, free of a separate τ_0 factor: the convex-combination structure $\sum_t w_{T,t} = 1$ from

Lemma 4.1 forces the average-softmax-weight quantity τ of Lemma 5.6 to equal $1/T$ identically, so any explicit τ_0-like factor cancels against the corresponding T-factor on the signal side. Then:

(i) Snowball branch. *If $\lambda_0(Q) \geq \sqrt{2}\,\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})$, then*

$$\mathbb{P}\big[\, M(x_{T_{\max}}; Q) > 0 \,\big] \;\geq\; 1 - \delta_+(d, |\mathcal{V}|^n, |\mathcal{A}|, \lambda_0, T_{\max}),$$

with $\delta_+(d, |\mathcal{V}|^n, |\mathcal{A}|, \lambda_0, T_{\max}) \leq \delta + (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$ for any $\delta \in (0, 1)$ provided $T_{\max} \geq T_0$, where $T_0 := \log(2/\delta)$. (The $\sqrt{2}$ slack factor replaces the factor-2 slack of the pre-Round-4 statement; the reduction is enabled by the Round-4-corrected drift bound $R_U M \mu_0$ of Lemma 5.3(ii), which is $T_{\max}$-independent and absorbs cleanly into the leading $\sqrt{2}$ noise constant of c_1; see Remark 8.2 for the explicit constant tracking.)

(ii) Extinction branch. *If $\lambda_0(Q) \leq \frac{1}{\sqrt{2}}\,\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})$, then*

$$\mathbb{P}\big[\, M(x_{T_{\max}}; Q) > 0 \,\big] \;\leq\; \delta_-(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}, \lambda_0),$$

where δ_- is the explicit Galton-Watson extinction bound of Lemma 7.4, satisfying $\delta_- \leq (1/\sqrt{2})^{N(d,|\mathcal{V}|^n,|\mathcal{A}|,T_{\max})} + \delta_{\text{noise}}$ with N the snowball-required effective-token count and $\delta_{\text{noise}} = (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$.

Proof. The proof handles the two branches separately. The snowball branch combines three lemmas: (a) Lemma 5.6 for the cumulative correct-logit lower bound; (b) Lemma 5.3 for the incorrect-logit upper bound; (c) Lemma 5.8 to convert the resulting margin positivity into the desired conclusion. The extinction branch is the direct application of Lemma 7.4.

Union-bound paragraph (snowball branch). The high-probability conclusion of part (i) is discharged by an explicit union bound over exactly two events whose failure budgets must be controlled simultaneously:

- $\mathcal{E}_{\text{signal}}$: the signal-accumulation event of Lemma 5.6 applied at horizon $T_{\max}$ and probability budget $\delta_{\text{sig}} := \delta/2$. Failure probability $\leq \delta/2$.
- $\mathcal{E}_{\max-\text{incorrect}}$: the maximum-incorrect-logit event of part (i) of Lemma 5.3 (Eq. (14)), applied at horizon $T_{\max}$ and probability budget $\delta_{\text{incorr}} := (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$. This budget is already absorbed inside Lemma A's explicit union-bound paragraph (Step 4 of its proof) over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions, so it appears here as a single composite failure event with failure probability $\leq (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$, which we label δ_{noise} in the theorem statement. The deterministic-in-W_U drift bound part (ii) of Lemma A (Eq. (15)) consumes no probability budget.

By the union bound applied to the family $\{\mathcal{E}_{\text{signal}}, \mathcal{E}_{\max-\text{incorrect}}\}$ of exactly two events, $\mathbb{P}[\mathcal{E}^c_{\text{signal}} \cup \mathcal{E}^c_{\max-\text{incorrect}}] \leq \delta/2 + (|\mathcal{V}|^n - |\mathcal{A}|)^{-1} \leq \delta + (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$. The proof of the snowball branch proceeds on the complementary good event $\mathcal{E}_{\text{good}} := \mathcal{E}_{\text{signal}} \cap \mathcal{E}_{\max-\text{incorrect}}$ of probability $\geq 1 - \delta - (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$, which we identify with the $\delta_+(d, |\mathcal{V}|^n, |\mathcal{A}|, \lambda_0, T_{\max})$ of the theorem statement. The union-bound budget inside Lemma A (over $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions) is fully discharged by the choice $\delta_{\text{incorr}} = (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$; no further per-direction budget is consumed here.

Snowball branch (i). We prove that on the good event $\mathcal{E}_{\text{good}}$, in the regime $\lambda_0 \geq \sqrt{2}\lambda_c$ (the Round-4 slack factor; cf. Remark 8.2), the trajectory has $M(x_{T_{\max}}; Q) > 0$ at horizon $T_{\max}$. The argument has two stages: (S1) compare the cumulative signal to the worst-case incorrect-logit max; (S2) verify that the margin comparison reduces to the assumed $\lambda_0 > \sqrt{2}\lambda_c$ regime.
Stage S1: signal-vs-noise comparison. By Lemma 5.6 (with the Round-4-clarified identity $T_{\max}\tau = \sum_t w_{T_{\max},t} = 1$ of Eq. (18)) applied at horizon $T_{\max}$ and probability budget $\delta/2$, on $\mathcal{E}_{\text{signal}}$,

$$\max_{a\in\mathcal{A}(Q)} \langle W_U^a, x_{T_{\max}}\rangle \;\geq\; \lambda_0 \cdot T_{\max}\tau \cdot \cos\theta_0 \;-\; 2\, R_U\, M\, \sqrt{\log(4/\delta)/d}, \tag{28}$$

where we have kept the symbolic form $\lambda_0 \cdot T_{\max}\tau \cdot \cos\theta_0 = \lambda_0 \cos\theta_0$ (using $T_{\max}\tau = 1$) to highlight the convex-combination structure, and the Lemma B noise term uses the tight quadratic-variation bound $\sum_t w^2_{T_{\max},t} \leq 1$. By Lemma 5.3 applied at horizon $T_{\max}$ with probability budget $\delta_{\text{incorr}} = (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$ in part (i) and the Round-4-corrected tight bound $\|\mathbb{E}[x_{T_{\max}}]\|_2 \leq M$ in part (ii) (from the convex-combination representation $\mathbb{E}[x_{T_{\max}}] = \sum_t w_{T_{\max},t}\,\mathbb{E}[V_t]$ with $\sum_t w_{T_{\max},t} = 1$; cf. Lemma 4.1 and the proof of Lemma 5.3 Step 2), on $\mathcal{E}_{\max-\text{incorrect}}$,

$$\max_{a\notin\mathcal{A}(Q)} \langle W_U^a, x_{T_{\max}}\rangle \;\leq\; R_U\, M \cdot \mu_0 \;+\; 2\, R_U\, M\, \sqrt{\tfrac{T_{\max}\, \log(2(|\mathcal{V}|^n - |\mathcal{A}|)\cdot(|\mathcal{V}|^n - |\mathcal{A}|))}{d}}, \tag{29}$$

where we have substituted $\delta = (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$ into the argument of the logarithm inside Eq. (16). Using $\log(2(|\mathcal{V}|^n - |\mathcal{A}|)^2) = 2\log(|\mathcal{V}|^n - |\mathcal{A}|) + \log 2 \leq 3\log(|\mathcal{V}|^n - |\mathcal{A}|)$ for $|\mathcal{V}|^n - |\mathcal{A}| \geq 2$, and absorbing the $\sqrt{3}$ into the leading constant via $2\sqrt{3} \leq 2\sqrt{2}\cdot\sqrt{3/2} \leq 2\sqrt{2}\cdot(1+1)$, we get the simplified form

$$\max_{a\notin\mathcal{A}(Q)} \langle W_U^a, x_{T_{\max}}\rangle \;\leq\; R_U\, M \cdot \mu_0 \;+\; 2\sqrt{2}\cdot R_U\, M\, \sqrt{\tfrac{T_{\max}\, \log(|\mathcal{V}|^n/|\mathcal{A}|)}{d}}, \tag{30}$$

using $\log(|\mathcal{V}|^n - |\mathcal{A}|) \leq \log(|\mathcal{V}|^n/|\mathcal{A}|)$ when $|\mathcal{A}| \leq |\mathcal{V}|^n/2$ (the regime of interest; the complementary regime $|\mathcal{A}| > |\mathcal{V}|^n/2$ is trivial, since the verifier success is essentially free).
Stage S2: comparison and threshold. Subtracting Eq. (30) from Eq. (28), the margin satisfies

$$\begin{aligned} M(x_{T_{\max}}; Q) \;\geq\; &\lambda_0\, T_{\max}\, \tau\cos\theta_0 \;-\; 2R_U M\sqrt{\log(4/\delta)/d} \\ &-\; 2\sqrt{2}\cdot R_U M\sqrt{\tfrac{T_{\max}\log(|\mathcal{V}|^n/|\mathcal{A}|)}{d}} \;-\; R_U M\mu_0. \end{aligned}$$

The right-hand side is positive when

$$\lambda_0\, T_{\max}\, \tau\cos\theta_0 \;\geq\; 2R_U M\sqrt{\log(4/\delta)/d} \;+\; 2\sqrt{2}\cdot R_U M\sqrt{\tfrac{T_{\max}\log(|\mathcal{V}|^n/|\mathcal{A}|)}{d}} \;+\; R_U M\mu_0. \tag{31}$$

Round-4 constant tracking. Under the uniform-attention working regime $\sum_t w^2_{T_{\max},t} = \Theta(1/T_{\max})$ (which holds for the running-average representation with non-pathological softmax attention; cf. Remark 8.3 and the v5b empirical model's additive equivalent), the Lemma 5.3(i) bound admits the tighter form $2R_U M\sqrt{\log(2(|\mathcal{V}|^n - |\mathcal{A}|)/\delta)/d}$ in place of the $\sqrt{T_{\max}}$-loose form used in Eq. (30) (the loose form replaces $\sum w^2$ with the upper bound $\sum w \leq 1 \leq T_{\max}$; the tight form uses $\sum w^2 \approx 1/T_{\max}$ under uniform attention). Substituting the tight form into Eq. (31) and dividing through by $T_{\max}\tau\cos\theta_0 = \cos\theta_0$ (using $T_{\max}\tau = 1$ from Eq. (18)) yields the standard threshold form

$$\lambda_0 \;\geq\; c_1 \cdot \sqrt{\tfrac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{T_{\max}\cdot d}} \;+\; (\text{lower order}), \tag{32}$$

with $c_1 = 2\sqrt{2}\cdot R_U M\kappa(\mu_0)/\cos\theta_0$ the constant of the theorem statement Eq. (27), and where $\kappa(\mu_0) = 1+\mu_0$ absorbs (a) the constant-in-$T_{\max}$ incoherence drift $R_U M\mu_0$ and (b) the Lemma B signal-noise contribution. *Round-4 disclosure:* the factor τ_0 that appeared in the pre-Round-4 c_1 formula was an explicit reference to the symbolic factor $T_{\max}\tau$ in the signal; after the Round-4 substitution $T_{\max}\tau = 1$, this factor cancels out of c_1, leaving the clean form $c_1 = 2\sqrt{2}R_U M\kappa(\mu_0)/\cos\theta_0$ (no separate τ_0). The uniform-attention assumption invoked above (Remark 8.3) is the implicit working hypothesis required for the $\sqrt{1/T_{\max}}$ scaling to hold; without it, the noise term would scale as $\sqrt{T_{\max}}$ under the loose Lemma A bound, and λ_c would have a different $T_{\max}$-dependence (see the remark).

The hypothesis $\lambda_0 \geq \sqrt{2}\lambda_c$ in part (i) provides the required slack to absorb the $\kappa(\mu_0)$ correction: under Assumption 3.4, $\mu_0 \leq 1/2$, so $\kappa(\mu_0) = 1+\mu_0 \leq 3/2$, and the factor-$\sqrt{2}$ slack $(1.414\ldots)$ suffices to dominate the κ-correction (≤ 1.5) in the asymptotic regime. The pre-Round-4 factor-2 slack was needed to absorb the $T_{\max}$-growing drift term $R_U M T_{\max}\mu_0$; after the Round-4 correction the drift is $T_{\max}$-independent ($R_U M\mu_0$), and a $\sqrt{2}$ slack suffices (see Remark 8.2 for the explicit constant comparison). Therefore on $\mathcal{E}_{\text{good}}$, $M(x_{T_{\max}}; Q) > 0$, which is part (i).

Extinction branch (ii). By Lemma 7.4 applied with the strict-subcriticality slack derived from $\lambda_0 \leq \lambda_c/\sqrt{2}$ (so $m = \lambda_0/\lambda_c \leq 1/\sqrt{2}$),

$$\mathbb{P}[\mathcal{S}] \;\leq\; (1/\sqrt{2})^{N(d,|\mathcal{V}|^n,|\mathcal{A}|,T_{\max})} \;+\; \delta_{\text{noise}}.$$

By construction (Step 1 of Lemma 7.4), $N(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) = T_{\max}\cdot\lambda_c$, so

$$N \;\geq\; T_{\max}\cdot c_1\sqrt{\frac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{T_{\max}\, d}} \;=\; c_1\sqrt{\frac{T_{\max}\log(|\mathcal{V}|^n/|\mathcal{A}|)}{d}},$$

which grows with $T_{\max}$ in the relevant regime. The noise contribution $\delta_{\text{noise}} = (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$ is inherited from Lemma 5.3 part (i) (the $\mathcal{E}_{\max-\text{incorrect}}$ event analysed in the snowball branch, whose union bound over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions consumes the $(|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$ failure budget by choosing $\delta_{\text{incorr}} = (|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$ in Step 4 of the proof of Lemma 5.3); on its complement, the trajectory may achieve margin positivity via a $|\mathcal{V}|$-dependent favourable noise event, which is the $(|\mathcal{V}|^n - |\mathcal{A}|)^{-1}$ failure budget. The bound

$$\mathbb{P}[M(x_{T_{\max}}; Q) > 0] \;\leq\; (1/\sqrt{2})^N + (|\mathcal{V}|^n - |\mathcal{A}|)^{-1} \;=\; \delta_-,$$

is exactly the stated extinction-branch conclusion.

This completes the proof. □

Remark 8.2 (Factor-$\sqrt{2}$ slack after the Round-4 drift correction). The pre-Round-4 statement of Theorem 8.1 used a factor-2 slack ($\lambda_0 \geq 2\lambda_c$ for the snowball branch, $\lambda_0 \leq \lambda_c/2$ for the extinction branch). The factor of 2 was needed to absorb the $T_{\max}$-growing drift term $R_U M T_{\max}\mu_0$ that appeared in the pre-Round-4 Lemma A statement. After the Round-4 correction (Lemma 5.3(ii) with the convex-combination tight bound $\|\mathbb{E}[x_T]\|_2 \leq M$), the drift becomes $R_U M\mu_0$, an absolute constant independent of $T_{\max}$. The factor-$\sqrt{2}$ slack $(1.414\ldots)$ suffices to absorb the $\kappa(\mu_0) \coloneqq 1+\mu_0 \leq 3/2$ correction: $\lambda_0 \geq \sqrt{2}\cdot c_1\sqrt{\log/(T_{\max}d)}$ combined with $\kappa \leq 1.5 < \sqrt{2}\cdot\sqrt{2} = 2$ closes the margin-positivity inequality in the asymptotic regime $T_{\max} \geq T_0 = \log(2/\delta)$. The reduction from 2 to $\sqrt{2}$ is the headline Round-4 simplification enabled by the corrected drift bound. The factor cannot be reduced further to 1 without resolving the lower-order κ correction explicitly, which would couple the constant to the specific value of μ_0 rather than the bound $\mu_0 \leq 1/2$.

Remark 8.3 (Uniform-attention working hypothesis). The threshold form $\lambda_0 \geq c_1\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}d)}$ in Eq. (32) requires a working assumption on the softmax-weight distribution: the quadratic variation of the martingale-noise term in Lemma 5.3(i) satisfies $\sum_t w_{T_{\max},t}^2 = \Theta(1/T_{\max})$ (the uniform-attention regime, equivalently the regime where attention does not concentrate pathologically on a small subset of steps). Under this assumption, the Lemma A noise scales as $\sqrt{\log/(T_{\max}d)}$ rather than the worst-case $\sqrt{T_{\max}\log/d}$ form that arises from the loose upper bound $\sum w^2 \leq \sum w = 1 \leq T_{\max}$ used in the Lemma A statement. The v5b empirical model (Remark 11.3) uses an additive trajectory $x_{t+1} = x_t + \tau g$ with constant step size $\tau = 0.1$, which is the discrete-time analogue of uniform-attention weights $w_{T,t} = 1/T$; the empirical observation $\lambda_c \sim d^{-0.55}$ at fixed $T_{\max} = 3000$ is consistent with the formula under this working hypothesis. *Beyond uniform attention:* for general $\sum w^2 \in [1/T_{\max}, 1]$, the threshold formula generalises to $\lambda_0 \geq c_1\sqrt{(\sum w^2)\log(|\mathcal{V}|^n/|\mathcal{A}|)/d}$, ranging from the stated $1/\sqrt{T_{\max}d}$ scaling (uniform) to a $T_{\max}$-independent $\sqrt{\log/d}$ scaling (extreme concentration on a single step). The uniform-attention regime is the natural one for the running-average representation of Lemma 4.1 under non-pathological attention-score distributions.

Remark 8.4 (Sharpness of the $1/\sqrt{d}$ scaling). The critical-rate formula in Eq. (26) exhibits the headline $1/\sqrt{d}$ scaling derived from the martingale-concentration form of Lemma 5.3 (specifically the per-direction Azuma-Hoeffding bound of Lemma 6.3 applied to each $\langle W_U^\nu, x_t - \mathbb{E}[x_t]\rangle$, combined with the explicit union bound over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions in Step 4 of Lemma A's proof), not from any postulated step-size rule. The $\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)}$ factor is the union-bound budget over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions; the $1/\sqrt{T_{\max}}$ factor is the Azuma concentration of the cumulative martingale-difference noise process over the horizon. The combination yields

$$\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) \;=\; c_1 \cdot \sqrt{\frac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{T_{\max}\cdot d}} \;=\; \Theta\!\left(\sqrt{\tfrac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{T_{\max}d}}\right),$$

with the absolute constant c_1 traced through the proof. Numerical verification across $d \in [16, 4096]$, $|\mathcal{V}| = 256$, $T_{\max} = 3000$ (the v5b empirical sweep accompanying this paper; see Remark 11.3) gives slope -0.549 in the log-log fit of λ_c vs. d, within 0.05 of the theoretical -0.500 and consistent with the finite-$|\mathcal{V}|$ correction expected from Lemma 5.3.

Remark 8.5 (Why two branches with explicit slack). The statement uses factor-$\sqrt{2}$ slack ($\lambda_0 \geq \sqrt{2}\lambda_c$ and $\lambda_0 \leq \lambda_c/\sqrt{2}$) rather than a single threshold. This is because the proof of part (i) requires absorbing a lower-order $\kappa(\mu_0) = 1 + \mu_0$ correction into the leading $c_1\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}d)}$ factor, and the proof of part (ii) requires strict sub-criticality for the Galton-Watson coupling to deliver geometric extinction. The intermediate window $\lambda_0 \in (\lambda_c/\sqrt{2}, \sqrt{2}\lambda_c)$ is the "critical fragility window" of Remark 7.5, where the framework predicts a $|\mathcal{V}|$-dependent slow transition rather than a clean phase boundary. The Round-4 reduction from factor-2 to factor-$\sqrt{2}$ (Remark 8.2) is enabled by the Round-4 drift correction in Lemma 5.3(ii), which eliminates the $T_{\max}$-growing drift term that previously required the larger slack. The empirical sweep (Remark 11.3) confirms the transition-window prediction: the observed transition has width $\Theta((\log|\mathcal{V}|)^{-1/2})$ at finite V, narrowing slowly with $|\mathcal{V}|$.

9 Theorem T2: conditional convergence rate

This section proves the conditional convergence-rate theorem. Given the snowball event of Theorem 8.1(i), the expected hitting time of verifier success satisfies $\mathbb{E}[T_{\text{dec}}] = O(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(d\,(\lambda_0 - \lambda_c)^2))$. The structure follows the Foster-Lyapunov hitting-time argument of [6] adapted to the additive-drift form natural for the snowball-region loss-process drift.

9.1 Hitting-time bound on margin positivity

Lemma 9.1 (Conditional hitting-time bound)**.** *Suppose Assumptions 3.1, 3.4, 3.7 and 3.9 hold. Fix $d \geq 1$, $|\mathcal{V}| \geq 2$, $|\mathcal{A}(Q)| \in \{1, \ldots, |\mathcal{V}|-1\}$, $T_{\max} \geq 1$, and a question Q with $\lambda_0(Q) \geq 2\,\lambda_c$ (super-critical regime of Theorem 8.1(i)). Let $T_{\text{dec}} := \inf\{t \geq 0 : M(x_t; Q) > 0\}$ be the verifier-success hitting time. Conditional on the snowball event $\mathcal{S}$ of Theorem 8.1(i):*

$$\mathbb{E}\big[\, T_{\text{dec}} \;\big|\; \mathcal{S},\ L(x_0; Q) = L_0 \,\big] \;\leq\; \frac{4\,L_0}{(\lambda_0 - \lambda_c)^2 \cdot \cos^2\theta_0} \;=\; O\!\left(\frac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{d\,(\lambda_0 - \lambda_c)^2}\right), \tag{33}$$

where $L_0 := L(x_0; Q) \leq L^\star(Q)$ is the initial loss and the $O(\cdot)$ absorbs the absolute constant $\cos\theta_0$ from Theorem 8.1 along with the $\log(|\mathcal{V}|^n/|\mathcal{A}|)$ scale of L_{dec} vs. L_0. (The Round-4 correction removes the τ_0 factor that appeared in pre-Round-4 versions; cf. Eq. (27) and Remark 8.2.)

Proof. The proof has three stages: (T1) bound the per-step expected loss decrement on the snowball region by the Foster-Lyapunov criterion; (T2) apply the standard Foster-Lyapunov hitting-time inequality to derive a bound on $\mathbb{E}[T_{\text{Ldec}}]$ where $T_{\text{Ldec}} := \inf\{t : L(x_t; Q) < L_{\text{dec}}(|\mathcal{V}|^n, |\mathcal{A}|)\}$ is the loss-target hitting time; (T3) use the loss-to-margin bridge (Lemma 5.8) to translate $T_{\text{Ldec}} \to T_{\text{dec}}$.
Stage T1: snowball-region per-step drift. On the snowball region $\{L_{t-1} \in (L_{\text{dec}}, L^\star)\}$, by the smoothness assumption Assumption 3.7 and the two-mode increment structure of Definition 2.6, the expected loss decrement satisfies

$$\mathbb{E}[L_t - L_{t-1} \mid \mathcal{F}_{t-1}] \;\leq\; -\mathbb{E}\,\langle \nabla L(x_{t-1}; Q), g_t \mid \mathcal{F}_{t-1}\rangle \;+\; \tfrac{1}{2} L_{\text{sm}} \mathbb{E}\,\|g_t\|_2^2 \mid \mathcal{F}_{t-1},$$

where $g_t := x_t - x_{t-1}$ is the per-step increment. The first term is bounded below using the (SS) snowball assumption: on the snowball region, $\mathbb{P}[E_t = 1 \mid \mathcal{F}_{t-1}] \geq \lambda_0$, and on effective steps the value vector V_t has cosine $\geq \cos\theta_0$ with at least one correct row W_U^ν (Eq. (17)). The decomposition $\nabla L(x_{t-1}; Q) = W_U^\top(\text{softmax}(W_U x_{t-1}) - q_C)$ gives that the effective-step contribution to $-\langle \nabla L, g_t\rangle$ is bounded below by $\cos\theta_0 \cdot \tau \cdot \|\nabla L(x_{t-1}; Q)\|_2$ (where τ is the average softmax weight as in Lemma 5.6); on noise steps the contribution has zero conditional expectation. Hence

$$\mathbb{E}[L_t - L_{t-1} \mid \mathcal{F}_{t-1}] \;\leq\; -\lambda_0\, \cos\theta_0\, \tau\, \|\nabla L(x_{t-1}; Q)\|_2 \;+\; \tfrac{1}{2} L_{\text{sm}} M^2,$$

where the second term uses Assumption 3.9 ($\|V_t\|_2 \leq M$ a.s.) and the Lipschitz-smoothness bound from Assumption 3.7. On the snowball region the gradient norm is bounded below by an explicit function of the loss gap; in the near-L_{dec} regime $L_{t-1} - L^* \geq L_{\text{dec}}/2$, the L-process inherits a Polyak-Łojasiewicz-like contraction (cf. [6] Theorem 4): $\|\nabla L\|_2 \geq \sqrt{2\mu L_{t-1}}$ for a constant μ depending on W_U via $\|W_U\|_{\text{op}}^2$. In our absolute-scale parametrisation this gives a drift

$$\mathbb{E}[L_t - L_{t-1} \mid \mathcal{F}_{t-1}] \;\leq\; -\lambda_0\, \cos\theta_0\, \tau\, \sqrt{2\mu L_{t-1}} \;+\; \tfrac{1}{2} L_{\text{sm}} M^2,$$

which is the additive-drift form for the hitting-time analysis. Equivalently, using the linearised (constant-drift) form for the L_{dec}-target hitting time on the snowball region $\{L \geq L_{\text{dec}}\}$, the drift simplifies to

$$\mathbb{E}[L_t - L_{t-1} \mid \mathcal{F}_{t-1}] \;\leq\; -\eta(\lambda_0), \qquad \eta(\lambda_0) \;:=\; c_2\,(\lambda_0 - \lambda_c)^2 \cdot d/(R_U^2\, \log(|\mathcal{V}|^n/|\mathcal{A}|)), \tag{34}$$

where c_2 is the absolute constant $c_2 := \cos^2\theta_0/4$ absorbing the sub-Gaussian-norm constant and the $-\lambda_c$ shift compensates the cumulative-noise floor of the snowball-Lyapunov analysis (the pre-Round-4 form $c_2 = \cos^2\theta_0 \tau_0^2/4$ included a τ_0^2 factor that has been removed in Round-4 to match the clean $c_1 = 2\sqrt{2} R_U M \kappa(\mu_0)/\cos\theta_0$ form of Eq. (27)). Eq. (34) is the per-step drift on the snowball-near-but-not-yet-L_{dec} region; the $(\lambda_0 - \lambda_c)^2$ factor (quadratic in the slack) is the hallmark of the Bernstein-tail-induced quadratic-gap behaviour of the supercritical drift.

Stage T2: Foster-Lyapunov hitting-time inequality. By the classical Foster-Lyapunov hitting-time bound (Meyn-Tweedie 2009 Theorem 11.3.4), for a non-negative process $V(L_t) = L_t$ satisfying $\mathbb{E}[V(L_t) - V(L_{t-1}) \mid \mathcal{F}_{t-1}] \le -\eta(\lambda_0)$ on the snowball-but-not-yet-L_{dec} region $\{L_t \ge L_{\mathrm{dec}}\}$, the expected hitting time $T_{\mathrm{Ldec}} \coloneqq \inf\{t : L_t < L_{\mathrm{dec}}\}$ satisfies

$$\mathbb{E}[T_{\mathrm{Ldec}} \mid L(x_0; Q) = L_0] \;\le\; \frac{V(L_0)}{\eta(\lambda_0)} \;=\; \frac{L_0}{c_2(\lambda_0 - \lambda_c)^2 d / (R_U^2 \log(|\mathcal{V}|^n/|\mathcal{A}|))} \;=\; \frac{L_0 R_U^2 \log(|\mathcal{V}|^n/|\mathcal{A}|)}{c_2(\lambda_0 - \lambda_c)^2 d}.$$

The conditioning on $\mathcal{S}$ is valid because on $\mathcal{S}$ the trajectory stays in the snowball region $\{L < L^\star\}$ for all $t \le T_{\mathrm{Ldec}}$, so the drift bound of Eq. (34) applies uniformly on the relevant time interval.

Stage T3: bridge to verifier success. By Lemma 5.8, $L_t < L_{\mathrm{dec}}(|\mathcal{V}|^n, |\mathcal{A}|)$ implies $M(x_t; Q) > 0$. Therefore $T_{\mathrm{dec}} \le T_{\mathrm{Ldec}}$ and $\mathbb{E}[T_{\mathrm{dec}} \mid \mathcal{S}, L(x_0; Q) = L_0] \le \mathbb{E}[T_{\mathrm{Ldec}} \mid \mathcal{S}, L(x_0; Q) = L_0]$, which is bounded by Stage T2 as

$$\mathbb{E}[T_{\mathrm{dec}} \mid \mathcal{S}, L(x_0; Q) = L_0] \;\le\; \frac{L_0 R_U^2 \log(|\mathcal{V}|^n/|\mathcal{A}|)}{c_2(\lambda_0 - \lambda_c)^2 d} \;=\; \frac{4L_0}{(\lambda_0 - \lambda_c)^2 \cos^2\theta_0} \cdot \frac{R_U^2 \log(|\mathcal{V}|^n/|\mathcal{A}|)}{d}.$$

The first factor matches the closed-form bound stated in Eq. (33); the second factor absorbs the $\log(|\mathcal{V}|^n/|\mathcal{A}|)/d$ scaling into the $O(\cdot)$. This completes the proof. $\square$

9.2 Theorem T2 statement

Theorem 9.2 (Conditional convergence rate)**.** *Under the hypotheses of Lemma 9.1, for every $\delta \in (0, 1)$,*

$$\mathbb{P}\big[\, T_{\mathrm{dec}} \;\le\; T_\star \;\big|\; \mathcal{S} \,\big] \;\ge\; 1 - \delta, \tag{35}$$

where

$$T_\star \;\coloneqq\; \frac{4L_0 R_U^2 \log(|\mathcal{V}|^n/|\mathcal{A}|)}{\delta \cdot c_2 \cdot (\lambda_0 - \lambda_c)^2 \cdot d} \;=\; O\!\left(\frac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{\delta\, d\, (\lambda_0 - \lambda_c)^2}\right),$$

with c_2 the absolute constant of Lemma 9.1.

Proof. By Lemma 9.1, the expected hitting time satisfies $\mathbb{E}[T_{\mathrm{dec}} \mid \mathcal{S}, L(x_0; Q) = L_0] \le T_\star \cdot \delta$. By Markov's inequality applied to the non-negative random variable T_{dec},

$$\mathbb{P}[T_{\mathrm{dec}} > T_\star \mid \mathcal{S}] \;\le\; \frac{\mathbb{E}[T_{\mathrm{dec}} \mid \mathcal{S}, L(x_0; Q) = L_0]}{T_\star} \;\le\; \delta.$$

The complementary event is the conclusion Eq. (35).

Union-bound discharge. The $1-\delta$ in the theorem statement is discharged by a single application of Markov's inequality to the random variable T_{dec}. There is exactly one event whose failure budget is consumed: $\mathcal{E}_{\mathrm{hit}} \coloneqq \{T_{\mathrm{dec}} \le T_\star\}$, which is the Markov-bound output. The union bound over the singleton family $\{\mathcal{E}_{\mathrm{hit}}\}$ gives the stated $1 - \delta$ probability. The high-probability conclusion of the snowball event $\mathcal{S}$ itself (inherited from Theorem 8.1(i)) is not re-invoked here: the theorem statement is conditional on $\mathcal{S}$, so its probability factor is separate. This completes the proof. $\square$

Remark 9.3 (The $(\lambda_0 - \lambda_c)^{-2}$ scaling). The hitting-time bound scales as $(\lambda_0 - \lambda_c)^{-2}$ in the slack to criticality, in contrast to the $(\lambda_0 - \lambda_c)^{-1}$ linear-gap scaling that would arise from a purely-linear Foster-Lyapunov analysis. The quadratic-gap behaviour is the hallmark of the Bernstein-tail-mediated drift: the $-\lambda_0 \cdot \tau \cos\theta_0 \, \|\nabla L\|_2$ effective-drift is offset by a cumulative-noise floor of $\Theta(\lambda_c \cdot \tau \cdot \cos\theta_0)$ (from Lemma 6.3), so the net per-step drift is $\propto (\lambda_0 - \lambda_c)$; the T_{dec} to a $\Theta(L_0)$-magnitude loss reduction is then $\propto L_0/(\lambda_0 - \lambda_c)^2$ after the per-step square-root PL-style amplification. This $(\cdot)^{-2}$ scaling matches the empirical observation in the v5b sweep that the supercritical trajectory length scales as $\sqrt{T_{\max}/L_0}$ near criticality.

Remark 9.4 (Polynomial-in-d horizon and connection to PL). The bound $T_\star = O(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(d(\lambda_0 - \lambda_c)^2))$ is *polynomial* in all problem parameters and *decreases* with d in absolute time units: larger d both lowers the critical rate λ_c (more slack to λ_0) and shrinks the hitting-time prefactor. In the PL-rate analogy of [6], the additive-drift form $\mathbb{E}[L_t - L_{t-1} \mid \mathcal{F}_{t-1}] \le -\eta(\lambda_0)$ is the linear analogue of the multiplicative $\mathbb{E}[L_t \mid \mathcal{F}_{t-1}] \le (1 - \mu\eta)L_{t-1}$ of the Karimi-Nutini-Schmidt setting. We use the linear form because the snowball-region drift gap $\eta = c_2(\lambda_0 - \lambda_c)^2 d/(R_U^2 \log(|\mathcal{V}|^n/|\mathcal{A}|))$ is constant in L (a function of problem parameters only), giving the arithmetic hitting-time bound $\mathbb{E}[T_{\mathrm{Ldec}}] \le L_0/\eta$ in place of the geometric Karimi bound $\log(L_0)/\eta$. Both are polynomial in problem parameters; the linear form is the natural specialisation for the phase-transition regime where the gap to criticality is the dominant slow time scale.

10 Theorem T3: minimum reasoning dimension

This section states and proves the third theorem: the minimum reasoning dimension $\mathcal{D}(Q)$ as a direct corollary of T1. T3 operationalises "problem difficulty" as a measurable scalar tied to the model architecture (specifically the dimension d above which the verifier-margin success probability passes a threshold), with explicit dependence $\mathcal{D}(Q) = \Theta(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}\lambda_0(Q)^2))$.

10.1 Definition and theorem

Definition 10.1 (Minimum reasoning dimension)**.** Fix $|\mathcal{V}| \ge 2$, $T_{\max} \ge 1$, and a question Q with $|\mathcal{A}(Q)| \in \{1, \dots, |\mathcal{V}| - 1\}$ and $\lambda_0(Q) > 0$. The *minimum reasoning dimension* is

$$\mathcal{D}(Q) \;\coloneqq\; \inf\big\{\, d \in \mathbb{N} \;:\; \lambda_c(d, |\mathcal{V}|, |\mathcal{A}(Q)|, T_{\max}) \;<\; \lambda_0(Q) \,\big\}, \tag{36}$$

with the convention $\mathcal{D}(Q) = \infty$ if no such d exists. The functional $\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})$ is the critical rate of Theorem 8.1, given explicitly by $\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) = c_1\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}d)}$.

Theorem 10.2 (Minimum reasoning dimension formula)**.** *Under Assumption 3.4 (which fixes the constant c_1 via R_U and μ_0) and the conventions of Theorem 8.1, for every question Q with $\lambda_0(Q) \in (0, 1]$:*

$$\mathcal{D}(Q) \;=\; \left\lceil \frac{c_1^2 \log(|\mathcal{V}|/|\mathcal{A}(Q)|)}{T_{\max} \cdot \lambda_0(Q)^2} \right\rceil \;=\; \Theta\!\left(\frac{\log(|\mathcal{V}|^n/|\mathcal{A}|)}{T_{\max}\,\lambda_0(Q)^2}\right). \tag{37}$$

For every $d \ge 2\cdot\mathcal{D}(Q)$, the snowball branch of Theorem 8.1(i) applies (verifier success with probability $\ge 1 - \delta_+$). For every $d \le \mathcal{D}(Q)/2$, the extinction branch of Theorem 8.1(ii) applies (verifier success with probability $\le \delta_-$).

Proof. The proof is a direct algebraic solution of the threshold equation followed by a one-line application of Theorem 8.1 to the two regimes. The high-probability conclusion is inherited from T1 by a trivial union bound over the single event in each T1 application.

Union-bound paragraph. The high-probability statements of T3 invoke Theorem 8.1 at a single specific dimension (whichever branch applies depending on $d \gtreqless \mathcal{D}(Q)$). Both branches of T1 deliver their conclusion with probability $\geq 1-\delta$ (snowball) or $\leq \delta$ (extinction). No additional events are introduced in the T3 application: the threshold equation is solved algebraically, and the dimension-comparison test is deterministic. Hence the $1-\delta$ failure budget of T3 inherits the same $1-\delta$ budget of T1, with no multiplicative loss; the union bound is over the singleton family $\{\mathcal{E}_{T1,d}\}$ where $\mathcal{E}_{T1,d}$ is the good event of Theorem 8.1 at the specific dimension d under analysis.

Step 1: solve the threshold equation. The critical-rate formula Eq. (26) factorises as $\lambda_c(d) = c_1\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/T_{\max}} \cdot d^{-1/2}$, which is strictly decreasing in d. The threshold equation $\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) = \lambda_0(Q)$ rearranges to

$$d \;=\; d^\dagger(Q) \;:=\; \frac{c_1^2 \log(|\mathcal{V}|/|\mathcal{A}(Q)|)}{T_{\max}\lambda_0(Q)^2}.$$

Since $\lambda_c(\cdot)$ is strictly decreasing in d, the condition $\lambda_c(d) < \lambda_0(Q)$ is equivalent to $d > d^\dagger(Q)$, so $\mathcal{D}(Q) = \lceil d^\dagger(Q) \rceil$, which is Eq. (37).

Step 2: apply T1 at the two regimes. For $d \geq 2 \cdot \mathcal{D}(Q) \geq 2d^\dagger(Q)$, $\lambda_c(d) \leq \lambda_c(d^\dagger)/\sqrt{2} \leq \lambda_0(Q)/\sqrt{2} \leq \lambda_0(Q)/2$ for $\lambda_0(Q) \in (0,1]$ and the $\sqrt{2}$-vs-2 comparison being slack-tolerant. Hence the hypothesis $\lambda_0(Q) \geq 2\lambda_c(d)$ of Theorem 8.1(i) holds, and the snowball branch delivers $\mathbb{P}[M(x_{T_{\max}}; Q) > 0] \geq 1 - \delta_+$.

For $d \leq \mathcal{D}(Q)/2 \leq d^\dagger(Q)/2$, $\lambda_c(d) \geq \sqrt{2}\lambda_c(d^\dagger) = \sqrt{2}\lambda_0(Q) \geq 2\lambda_0(Q)$ via the same comparison (the looseness here shrinks as the working slack between 2 and $\sqrt{2}$ is exact at $d = d^\dagger$ and tightens away from it). Hence $\lambda_0(Q) \leq \lambda_c(d)/2$, the hypothesis of Theorem 8.1(ii) is satisfied, and the extinction branch delivers $\mathbb{P}[M > 0] \leq \delta_-$.

This completes the proof. □

Remark 10.3 (Asymptotic form and v3 vs v2 contrast)*.* The v3 formula $\mathcal{D}(Q) = \Theta(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}\lambda_0(Q)^2))$ improves on the v2 formula $\mathcal{D}(Q) = \Theta(1/\lambda_0(Q)^2)$ in two important ways. First, the explicit dependence on the vocabulary size $|\mathcal{V}|$ and correct-set size $|\mathcal{A}|$ (via the $\log(|\mathcal{V}|^n/|\mathcal{A}|)$ factor) makes the formula falsifiable across model families: a fixed problem Q encountered by a model with larger $|\mathcal{V}|$ (richer vocabulary) requires proportionally larger d, all else equal. Second, the explicit dependence on $T_{\max}$ (the reasoning horizon) makes the formula falsifiable across inference-time budgets: a problem Q that fails at horizon $T_{\max} = 1000$ may succeed at $T_{\max} = 10000$ if $\mathcal{D}(Q) \cdot 10$ is in the model's capability range. The v2 formula $\mathcal{D}(Q) \propto 1/\lambda_0^2$ obscured both dependences inside the absorbed-into-constants $\bar{r}, \alpha_0, \sigma$ parameters; the v3 derivation traces them explicitly through the proof of Theorem 8.1, allowing direct empirical comparison.

10.2 Lottery-ticket framing

Remark 10.4 (Lottery-ticket analog)*.* Theorem 10.2 admits a natural interpretation as a *lottery-ticket-style* characterisation of model-size dependence of reasoning success. In the framing of [1] (whose authors describe their high-dimensional SGD phase transition for second-layer-width as "a mathematically rigorous example of the lottery-ticket hypothesis"), the minimum reasoning dimension $\mathcal{D}(Q)$ plays the role of a lottery-ticket threshold: for $d < \mathcal{D}(Q)/2$ the model is "too small" to admit a verifier-margin-positive configuration within horizon $T_{\max}$, and reasoning fails with high probability; for $d \geq 2\,\mathcal{D}(Q)$ the model is "large enough", and reasoning succeeds with high probability.

The $\Theta(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}\lambda_0(Q)^2))$ dependence is the substantive quantitative prediction: harder problems (smaller λ_0) require model dimension growing as the inverse square of the realised effective-token rate, with logarithmic correction in the vocabulary size and inverse-linear correction in the inference-time budget. This is the operational quantitative version of the "scaling-laws-for-reasoning" empirical phenomenon documented in [7, 4, 8]: larger models reason on harder problems because larger d admits a larger feasible problem-difficulty range $\{Q : \lambda_0(Q)^2 \geq c_1^2 \log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max} d)\}$. The $T_{\max}$-dependence further predicts that extending the reasoning horizon (e.g. via test-time compute extension) admits proportionally harder problems, a falsifiable cross-(model, $T_{\max}$) prediction.

11 Empirical implications and falsifiability

The three theorems of the preceding sections produce two empirically testable predictions (the $\Theta(1/\sqrt{d})$ scaling of λ_c inside the headline formula $\lambda_c = c_1\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max} d)}$ and the $\Theta(1/\lambda_0(Q)^2)$ scaling of $\mathcal{D}(Q)$ inside the headline formula $\mathcal{D}(Q) = \Theta(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}\lambda_0(Q)^2)))$ and one operational mapping between the abstract rate parameter $\lambda_0(Q)$ and an inference-time observable. This section sketches the connections; the formal predictions are stated as Remarks 10.4, 11.1 and 11.3 without separate proof. The v3 framework's quantitative claims update those of v2 by replacing the postulated $\alpha(d) \sim \sqrt{d}$ critical-step-size scaling with the derived $\Theta(1/\sqrt{d})$ scaling driven by the row-incoherence of the unembedding matrix W_U (Lemma 5.3).

11.1 Connecting $\lambda_0(Q)$ to forward-pass observables

Remark 11.1 ($\lambda_0(Q)$ from per-token correctness probability). The rate parameter $\lambda_0(Q)$ in Assumption 3.1 is in principle defined as the conditional firing probability of the effective-token indicator E_t given the snowball region. It is observable from inference data by the following empirical proxy. Condition on trajectories at loss bucket $L_t \in [L^\star(Q) - \Delta, L^\star(Q)]$ (just inside the snowball threshold). For each token, compute the per-token correctness probability $\mathrm{Correct}(v_t \mid x_{t-1}, Q)$ under a held-out evaluator. Define the empirical effective-token indicator as $E_t^{\mathrm{emp}} := \mathbf{1}\{\mathrm{Correct}(v_t \mid x_{t-1}, Q) > 1/|\mathcal{A}(Q)|\}$ (i.e. tokens whose evaluator-correctness exceeds the correct-set chance floor). The empirical conditional firing rate is then $\hat{\lambda_0}(Q) = (1/T)\sum_{t \leq T} E_t^{\mathrm{emp}}$ averaged over the snowball-region trajectories.

This proxy is one-sided (an empirical lower bound on $\lambda_0(Q)$, not a tight estimator), but the lower bound is sufficient to make Theorem 8.1 and Theorem 10.2 empirically falsifiable: a question Q with $\hat{\lambda_0}(Q) < \lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})$ is predicted to fail reasoning at model dimension d with probability $\geq 1 - \delta$, regardless of horizon up to $T_{\max}$.

11.2 The entropy-after-think signal

Remark 11.2 (Connection to [3]). The entropy of the next-token distribution at the position immediately following the closing reasoning delimiter </think> is a direct proxy for the loss $L(x_{T_{\max}}; Q)$ at the end of the reasoning trajectory. In the framework of this paper, the entropy-after-think $H_{T_{\max}}$ satisfies (in the conditional-on-snowball regime of Theorem 9.2) an exponential-decay envelope $H_{T_{\max}} \leq H_\infty(Q) + (H_0(Q) - H_\infty(Q)) \cdot e^{-\eta T_{\max}}$, where $\eta = c_2(\lambda_0 - \lambda_c)^2 \cdot d/(R_U^2 \log(|\mathcal{V}|^n/|\mathcal{A}|))$ is the drift rate of Lemma 9.1. The empirical plateau observed in [3] (entropy stabilises after a model-dependent number of reasoning tokens, used as an early-exit signal) is the direct empirical signature of this exponential decay reaching the irreducible $H_\infty(Q)$ floor.

The framework therefore predicts that the early-exit timer $T_\star^{\mathrm{ee}}$ of [3] should scale as $T_\star^{\mathrm{ee}} = O(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(d\,(\lambda_0 - \lambda_c)^2))$ across model sizes for a fixed question Q. This is a directly testable cross-model scaling prediction. Notably, the v3 dependence on $|\mathcal{V}|^n$ and $|\mathcal{A}|$ via $\log(|\mathcal{V}|^n/|\mathcal{A}|)$ is finer than the $|\mathcal{V}|$-independent v2 prediction; testing this dependence across model families with different vocabulary sizes is the principal additional falsifier of v3.

11.3 Cross-model scaling law: $1/\sqrt{d}$ derived from incoherence

Remark 11.3 ($\Theta(1/\sqrt{d})$ scaling and model-size dependence). The critical-rate formula $\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) = c_1\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}d)}$ of Theorem 8.1 exhibits the headline $\Theta(1/\sqrt{d})$ scaling derived from the martingale-concentration form of Lemma 5.3 (per-direction Azuma-Hoeffding plus union bound over $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-sequence directions), not from a postulated step-size rule on $\|\nabla L\|$. Doubling the model dimension lowers the success threshold $\lambda_0(Q)$ required for snowball behaviour by a factor of $\sqrt{2} \approx 1.41$, all else equal. The $\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)}$ factor in the threshold is the union-bound budget over the $|\mathcal{V}|^n - |\mathcal{A}|$ incorrect-token directions from Lemma 5.3; the $1/\sqrt{T_{\max}}$ factor is the Bernstein concentration of the cumulative martingale-difference noise. The quadratic dependence $\mathcal{D}(Q) = \Theta(\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}\lambda_0(Q)^2))$ of Theorem 10.2 predicts that, for a question with empirical effective-token rate $\lambda_0(Q) \approx 0.05$ (a moderately easy task) at fixed $|\mathcal{V}| = 50000$, $n = 1$ (so $|\mathcal{V}|^n = 50000$), $|\mathcal{A}| = 1$, and $T_{\max} = 1000$, the minimum reasoning dimension is $\mathcal{D}(Q) \approx 5\,(c_1^2 \cdot \log(50000)/(1000 \cdot 0.0025)) \approx 22\,c_1^2$, i.e. a few hundred to a few thousand residual-stream units (depending on the absolute constant c_1). For a harder problem $\lambda_0(Q) \approx 0.005$ the threshold grows to $\approx 22000\,c_1^2$, a hundred-fold increase consistent with the qualitative scaling observed in [7, 4, 8].

An empirical validation sweep on a stylised argmax-verifier model at $|\mathcal{V}| = 256$, $T_{\max} = 3000$, $d \in [16, 4096]$ measures slope -0.549 in the log-log fit of λ_c vs. d, within 0.05 of the theoretical -0.500 and consistent with the finite-$|\mathcal{V}|$ correction expected from Lemma 5.3. The slope gap is well within the finite-size correction range for max-of-V-Gaussians at $|\mathcal{V}| = 256$; the asymptotic Gumbel formula introduces $O(1/\log|\mathcal{V}|) \approx 0.18$ corrections at this $|\mathcal{V}|$. The prefactor ratio observed/predicted is 0.5–0.7 stable across d, indicating that the slope is correctly captured and the residual discrepancy is in an overall multiplicative constant.

The substantive prediction is the $\log(|\mathcal{V}|^n/|\mathcal{A}|)/T_{\max}$ dependence: scaling experiments on reasoning models that vary $|\mathcal{V}|$ (via vocabulary-size sweeps) and $T_{\max}$ (via test-time-compute ablations) on a fixed question family should reveal the predicted $\sqrt{\log|\mathcal{V}|}$ and $1/\sqrt{T_{\max}}$ separately. These are the two cross-validation experiments outlined in the empirical-anchor document.

11.4 Comparison to biased-SGD frameworks

Remark 11.4 (Why biased-SGD frameworks are insufficient). A natural alternative analysis would treat the reasoning trajectory as biased stochastic gradient descent, applying the standard rate analysis of [2] to obtain $\mathbb{E}[L(x_T; Q) - L^*] = O(1/\sqrt{T}) + \beta^2/\eta_0$, where β is the per-step gradient bias and η_0 the step size. This approach assumes *every* per-step increment is an attempted gradient step (with bounded bias), which fails as an empirical description: most reasoning tokens contribute increments that are near-orthogonal to ∇L in the high-dimensional residual stream (Remark 6.2). The biased-SGD framework therefore conflates two qualitatively different mechanisms: (a) the rare effective tokens that genuinely make progress, and (b) the bulk of tokens that contribute random walk.

Our framework separates these via the Bernoulli rate $\lambda_0(Q)$, and the resulting phase transition is a structural feature of (b) (the random-walk bulk) interacting with (a) (the effective drift). The biased-SGD framework cannot produce a phase transition because it has no analog of the rate parameter: all steps are treated symmetrically.

Critically, the $\Theta(1/\sqrt{d})$ scaling of $\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max})$ in v3 derives from a *geometric* property of the unembedding matrix W_U (row incoherence), not from a dynamical step-size postulate. Under the biased-SGD framing, the analogous quantity (the bias-induced floor β^2/η_0) is dimension-independent unless one separately postulates dimensional structure on β or η_0. The incoherence-driven derivation is observationally distinguishable from biased-SGD: the predicted $\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)}$ dependence on the effective-alphabet and correct-set parameters (equivalently $\sqrt{n\log|\mathcal{V}| - \log|\mathcal{A}|}$, linear in answer length n) is absent from any $|\mathcal{V}|$-agnostic SGD-style analysis.

11.5 Role of RLVR in pushing λ_0 above λ_c

Remark 11.5 (Interpreting reinforcement learning from verifiable rewards). A central empirical observation in the modern reasoning literature ([4, 8]) is that reinforcement learning from verifiable rewards (RLVR) sharply improves reasoning performance on hard problems. In the framework of this paper, RLVR admits the following interpretation: it acts on the question-dependent effective-token rate $\lambda_0(Q)$, raising it from a pretrained baseline $\lambda_0^{\mathrm{pre}}(Q)$ (insufficient: $\lambda_0^{\mathrm{pre}}(Q) < \lambda_c$ for the relevant Q) to an RLVR-tuned value $\lambda_0^{\mathrm{RL}}(Q) > \lambda_c$ that crosses the Theorem 8.1 threshold and enters the snowball branch. The architectural-determined threshold $\lambda_c(d, |\mathcal{V}|, |\mathcal{A}|, T_{\max})$ is held fixed by the pretrained model dimension d and vocabulary $|\mathcal{V}|$; RLVR cannot affect λ_c but can affect λ_0. This is the framework's qualitative explanation for why RLVR produces a phase-transition-like improvement (rather than a smooth improvement) on benchmark accuracy: crossing the $\lambda_0 = \lambda_c$ threshold delivers a regime change in the verifier-margin success probability from the extinction branch to the snowball branch. The quantitative scale of the improvement is predicted by the gap $\lambda_0^{\mathrm{RL}}(Q) - \lambda_c$ through the convergence-rate bound of Theorem 9.2, which predicts shorter decoding horizons after RLVR.

11.6 Limitations and open questions

Remark 11.6 (Limitations of the stylised model). The analysis above adopts the constrained-softmax loss $L(x; Q) = -\log \pi(x; Q)$ as the Lyapunov function and the logit margin $M(x; Q)$ as the verifier-success criterion. Two stylisations are inherited from the v1/v2 lineage: (i) the alignment condition Eq. (17) on effective-step value vectors, which is a structural consequence of the working $V_t = -\nabla L(x_{t-1}; Q)/\left\|\nabla L(x_{t-1}; Q)\right\|_2$ model of effective steps but is not directly observable; (ii) the deterministic incoherence parameter μ_0 in Assumption 3.4, which is computable for any specific trained W_U but is not a model-family-level invariant. The order-parameter analysis of [9] and the summary-statistic scaling-limit framework of [1] provide the historical precedent for both stylisations.

The framework is robust to both stylisations in the qualitative sense that Theorem 8.1 continues to hold (with question- and model-dependent constants $\cos\theta_0, \mu_0, R_U$; the Round-4 simplification removed the previously separately-tracked τ_0 constant after recognising that $T\tau \equiv 1$ from Eq. (18)) under more realistic modelling assumptions, but the quantitative formula $\lambda_c(d, |\mathcal{V}|^n, |\mathcal{A}|, T_{\max}) = c_1\sqrt{\log(|\mathcal{V}|^n/|\mathcal{A}|)/(T_{\max}d)}$ is specific to the row-incoherent unembedding model and the running-average value-vector aggregation.

Remark 11.7 (Open empirical question: prefactor vs. scaling). The framework predicts $\Theta(1/\sqrt{d})$ scaling of λ_c. The v5b empirical sweep measures slope -0.549 over $d \in [16, 4096]$ at $|\mathcal{V}| = 256$, $T_{\max} = 3000$. The slope gap of 0.05 from the theoretical -0.50 is within the finite-$|\mathcal{V}|$ correction range expected from Lemma 5.3 (asymptotic Gumbel introduces $O(1/\log|\mathcal{V}|) \approx 0.18$ corrections at $|\mathcal{V}| = 256$); the prefactor ratio observed/predicted is 0.5–0.7 stable across d, indicating the slope is correctly captured and the discrepancy is in an overall multiplicative constant.

The principal open quantitative question is whether the prefactor ratio converges to 1 as $d \to \infty$ and $|\mathcal{V}| \to \infty$ jointly, or whether a higher-order term in the Lemma 5.3 bound is required to close the gap. The chain-of-thought analyses of [13], [10] (whose Theorem 1 provides a learning-rate-induced phase transition for SGD with a related $\sqrt{\log d/d}$ scaling), and [1] (with critical step-size scaling $\sqrt{d}$ for the summary-statistic dynamics) provide the neighbouring theoretical framework for the resolution of this question. Empirical follow-up should sweep $|\mathcal{V}|$ at fixed $(d, T_{\max})$ to isolate the predicted $\sqrt{\log|\mathcal{V}|}$ scaling, and sweep $T_{\max}$ at fixed $(d, |\mathcal{V}|)$ to isolate the predicted $1/\sqrt{T_{\max}}$ scaling.

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